UNIT

I

ENGINEERING PHYSICS • DIGITAL NOTES

Wave Optics

Interference • Thin films • Newton's rings • Diffraction • Grating • Polarisation • Nicol prism • Wave plates

QUICK SUMMARY • What you will learn

- Why light waves add up to bright and dark fringes, and how thin films and Newton's rings use this
- How light bends around obstacles: single slit, double slit and the diffraction grating
- How light can be "combed" into one direction of vibration (polarisation) and where we use it



Fig. 1: Roadmap of Unit I: every topic follows from the wave nature of light

1 Interference of Light

Principle of superposition: when two or more waves meet at a point, the resultant displacement is the **vector sum** of the individual displacements. Waves pass through each other unchanged and separate again.

DEFINITION • Interference

The redistribution of light energy in space when two coherent waves superpose, giving alternate bright (maxima) and dark (minima) regions.

	Constructive interference	Destructive interference
Phase difference φ	0, 2π, 4π ... = 2nπ	π, 3π ... = (2n+1)π
Path difference Δ	nλ	(2n+1)λ/2
Resultant amplitude	$a_1 + a_2$ (maximum)	$\lvert a_1 - a_2 \rvert$ (minimum)
Intensity	$(a_1 + a_2)^2$	$(a_1 - a_2)^2$ (zero if $a_1 = a_2$)

Resultant intensity of two waves

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

Phase ↔ path difference

$$\phi = \frac{2\pi}{\lambda}\Delta$$

REMEMBER • Conditions for sustained interference

- The sources must be **coherent** (constant phase difference, same frequency).
- Sources should be **monochromatic** and of nearly equal amplitude (for good contrast).
- Sources must be very close together/narrow; light is obtained from **one source split into two** (by division of wavefront or division of amplitude).
- Energy is conserved: the bright fringes gain exactly what the dark fringes lose.

Two ways of getting coherent sources

Method	Idea	Examples
Division of wavefront	Different parts of the same wavefront act as two sources	Young's double slit, Fresnel biprism
Division of amplitude	One beam is partly reflected and partly transmitted	Thin films, Newton's rings, Michelson interferometer

Young's fringe width (slit gap d, screen distance D)

$$\beta = \frac{\lambda D}{d}$$

2 Interference in Thin Films (Reflection Geometry)

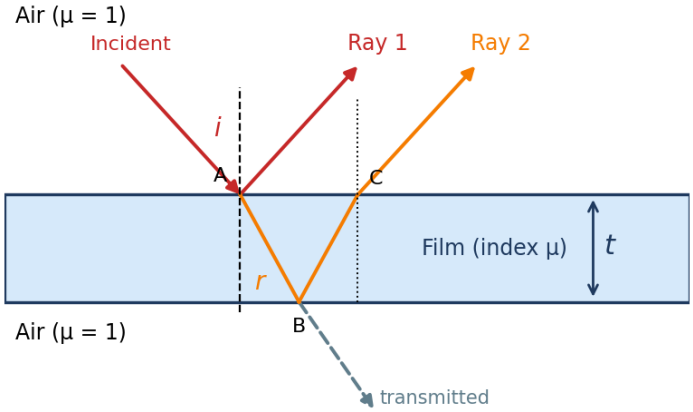


Fig. 2: A plane-parallel film of thickness t. Rays 1 and 2 come from the same incident ray and are coherent (division of amplitude).

Ray 1 reflects from the top surface (air → denser film), so it suffers a **phase change of π** (equivalent to λ/2). Ray 2 reflects at the bottom surface (film → air), where there is no phase change. Using geometry (Snell's law, μ = sin i / sin r):

Optical path difference (reflected)	$\Delta = 2\mu t\cos r \ \left(+\frac{\lambda}{2}\right)$
Bright film (maxima)	$2\mu t\cos r = (2n+1)\frac{\lambda}{2} \qquad n = 0, 1, 2, \ldots$
Dark film (minima)	$2\mu t\cos r = n\lambda \qquad n = 0, 1, 2, \ldots$

KEY IDEA • Transmitted light is complementary

In transmission, the conditions swap: bright when 2μt cos r = nλ and dark when 2μt cos r = (2n+1)λ/2. Where reflection shows a bright band, transmission shows a dark one.

Colours in thin films

- White light has many wavelengths. For a given thickness t and angle r, **only some wavelengths satisfy the bright condition**, so the film looks coloured (soap bubbles, oil on water, peacock feathers).
- Colour changes with thickness and viewing angle, because Δ = 2μt cos r changes.
- A very thin film (t ≪ λ) looks **black** in reflected light, as Δ ≈ λ/2 gives destructive interference (black spot on a bubble just before it bursts).
- A thick film shows no colours: too many orders overlap, so the eye sees white.

APPLICATIONS • Applications of thin-film interference

- **Anti-reflection (AR) coating:** a film of thickness t = λ/4μ and index μ ≈ $\sqrt{(\mu_{glass})}$ (MgF_2, μ = 1.38) cancels reflected light, so lenses, solar cells and spectacles transmit more light.
- **Highly reflecting coatings** (multilayer λ/4 stacks) for laser mirrors and dielectric mirrors.
- **Interference filters** that pass one narrow wavelength band.
- Testing the **flatness** of optical surfaces and measuring very small thicknesses.

EXAMPLE • Soap film

Find the minimum thickness of a soap film (μ = 1.33) that looks bright in reflected light of λ = 5890 Å at normal incidence.

2μt = λ/2 (n = 0) → t = λ/4μ = 5890×10^{-10} / (4 × 1.33) = **1.107×10^{-7} m (≈ 111 nm)**.

3 Newton's Rings

A plano-convex lens of large radius of curvature R is placed on an optically flat glass plate. A thin **air film of increasing thickness** forms between them. Monochromatic light falls normally (via a 45° glass plate G) and the interference of rays reflected from the top and bottom of the air film gives **concentric circular fringes** seen through the microscope M.

- Centre (t = 0): path difference is λ/2 (due to the π phase change), so the centre is **dark**.
- Rings get closer together as we move outward (t grows as r^2).
- Fringes are localised in the air film.

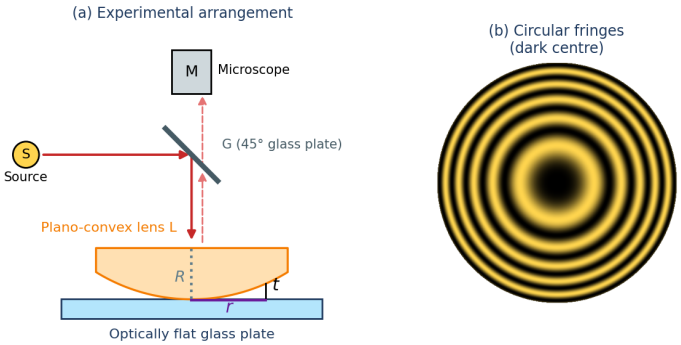


Fig. 3: Newton's rings set-up and pattern

Theory (reflected light, normal incidence, air film μ = 1)

For a ring of radius r at film thickness t: from the right triangle in the lens, $R^2 = r^2 + (R - t)^2 \rightarrow r^2 = 2Rt - t^2 \approx 2Rt$ (as $t \ll R$), hence $t = r^2/2R = D^2/8R$. The path difference is $2t + \lambda/2$.

Dark rings: $2t = n\lambda$	$D_n^2 = 4\,n\lambda R \qquad (D_n \propto \sqrt{n})$
Bright rings: $2t = (2n-1)\lambda/2$	$D_n^2 = 2\,(2n-1)\,\lambda R$
Wavelength of light	$\lambda = \dfrac{D_{n+p}^2 - D_n^2}{4\,p\,R}$
Refractive index of a liquid	$\mu = \dfrac{D_{n+p}^2 - D_n^2\ (\text{air})}{D_{n+p}^2 - D_n^2\ (\text{liquid})}$

KEY IDEA • Procedure

- **Wavelength:** measure the diameters of the n-th and (n+p)-th dark rings with a travelling microscope, using the known R; plot D^2 against n (straight line, slope = 4λR).
- **Refractive index:** fill the space between lens and plate with a liquid. The ring diameters shrink because the path difference becomes 2μt; $D^2 \rightarrow D^2/\mu$, so μ = (slope in air)/(slope in liquid).

EXAMPLE • Numerical

The diameters of the 4th and 12th dark rings are 0.400 cm and 0.700 cm. If λ = 5893 Å, find R.
$R = (D_{12}^2 - D_4^2) / (4\,p\,\lambda) = (4.9\times10^{-5} - 1.6\times10^{-5}) / (4 \times 8 \times 5.893\times10^{-7}) \approx$ **1.75 m**.

APPLICATIONS • Uses of Newton's rings

- Measuring the wavelength of monochromatic light and the refractive index of liquids.
- Finding the radius of curvature of a lens (R known → λ; λ known → R).
- Testing how flat or spherical a lens surface is: irregular rings show a defect.

REMEMBER • Common mistakes

- Use **diameter**, not radius, in $D^2 = 4n\lambda R$.
- Count rings from the **first dark ring** (n = 1); the centre is n = 0.
- Convert units: cm → m and Å → m before substituting.

4 Diffraction

DEFINITION • Diffraction

The bending of light around the edges of an obstacle or aperture and its spreading into the geometrical shadow. It is appreciable when the aperture size is comparable to λ.

Feature	Fresnel diffraction	Fraunhofer diffraction
Source / screen distance	Finite (near field)	Infinite (far field), usually using lenses
Wavefront	Spherical or cylindrical	Plane
Lenses needed?	No	Yes (collimating and focusing)
Analysis	Complicated (Fresnel zones)	Simple (Fourier-type)

Interference	Diffraction
Superposition of waves from two (or few) coherent sources	Superposition of secondary wavelets from one wavefront
All bright fringes are equally bright (ideal case)	Intensity falls rapidly for higher orders
Fringes usually equally spaced; dark fringes are nearly black	Fringes unequally spaced; minima are not perfectly dark

Fraunhofer diffraction at a single slit (width a)

The slit is divided into a large number of secondary sources; waves from different parts interfere. The path difference between waves from the two edges is a sinθ.

KEY IDEA • Why minima occur at a sinθ = nλ

- Split the slit into two halves. Every ray in the top half has a partner in the bottom half a distance a/2 away.
- If a sinθ = λ, the path difference of each pair is λ/2, so all pairs cancel and the intensity is zero. The same idea with 4, 6 ... parts gives nλ.
- Maxima lie roughly midway: a sinθ ≈ (2n+1)λ/2.

Intensity	$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2, \qquad \alpha = \frac{\pi a \sin\theta}{\lambda}$
Minima (α = nπ)	$a\sin\theta = n\lambda, \quad n = \pm 1, \pm 2, \ldots$
Secondary maxima	$\tan\alpha = \alpha \;\; (\alpha \approx 1.43\pi,\ 2.46\pi, \ldots)$

APPLICATIONS • Applications of diffraction

- Grating spectrometers to study the spectra of stars and materials.
- X-ray diffraction to find crystal structure (Unit II).
- CD/DVD surfaces show rainbow colours; limits the resolving power of telescopes and microscopes.

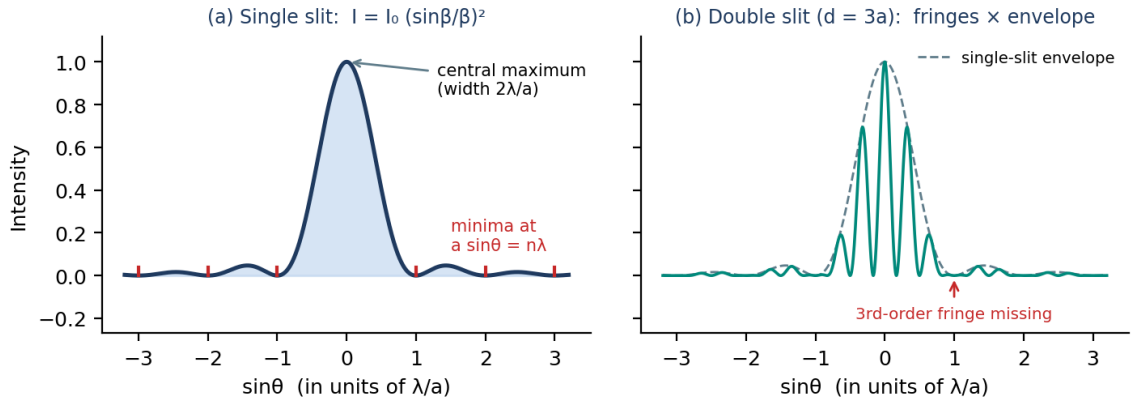


Fig. 4: (a) single-slit pattern, (b) double-slit pattern = cos² fringes inside a single-slit envelope

- Central maximum is the brightest and twice as wide as the others (angular width 2λ/a); secondary maxima fall as 1 : 0.045 : 0.016 ...
- Narrower slit (smaller a) gives a **wider** pattern.

5 Double Slit, N-Slits and the Diffraction Grating

Double slit (slit width a, separation d): the intensity is the product of the single-slit diffraction envelope and the two-source interference term.

Double slit intensity	$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2\cos^2\beta, \qquad \beta = \frac{\pi d\sin\theta}{\lambda}$
Interference maxima	$d\sin\theta = n\lambda$
Missing orders if d/a = m	$n = m,\ 2m,\ 3m, \ldots$

N slits (qualitative): the pattern gets an extra factor (sin Nβ / sin β)². Principal maxima stay at d sinθ = nλ but become **sharper and brighter (intensity ∝ N²)**. Between two principal maxima there are **(N−1) minima** and **(N−2) weak secondary maxima**.

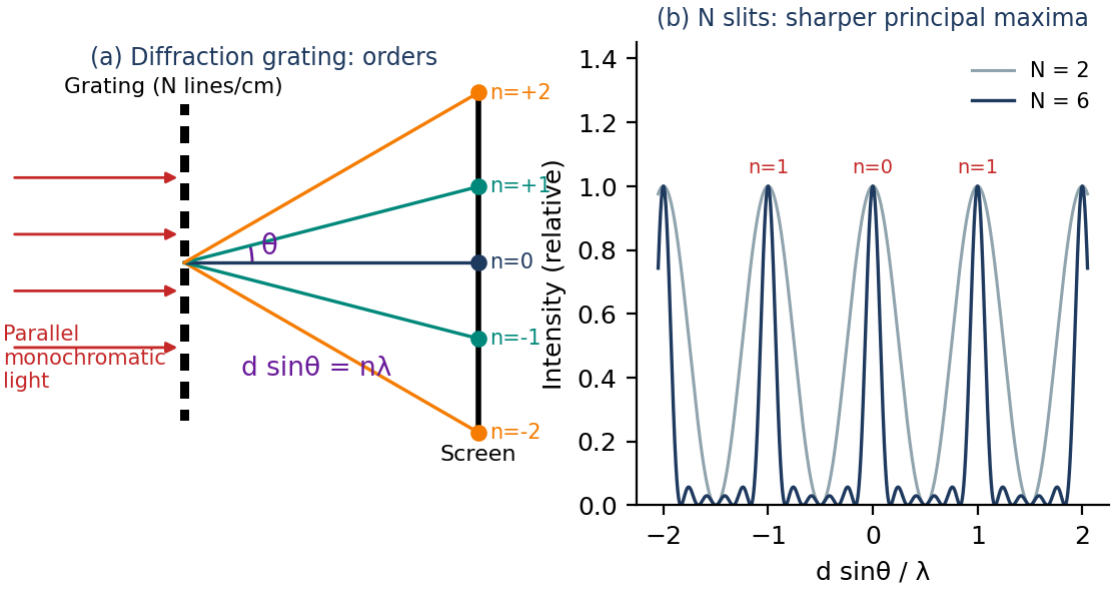


Fig. 5: (a) orders from a grating; (b) more slits → narrower principal maxima

DEFINITION • Diffraction grating

A glass or metal plate with a very large number of equally spaced, parallel, narrow slits (about 6000 lines/cm). Grating element (a + b) = 1/N, with a = slit width and b = opaque width.

Grating equation	$(a+b)\sin\theta_n = n\lambda$
Highest possible order	$n_{max} < \frac{a+b}{\lambda}$

Dispersive power and resolving power (qualitative)

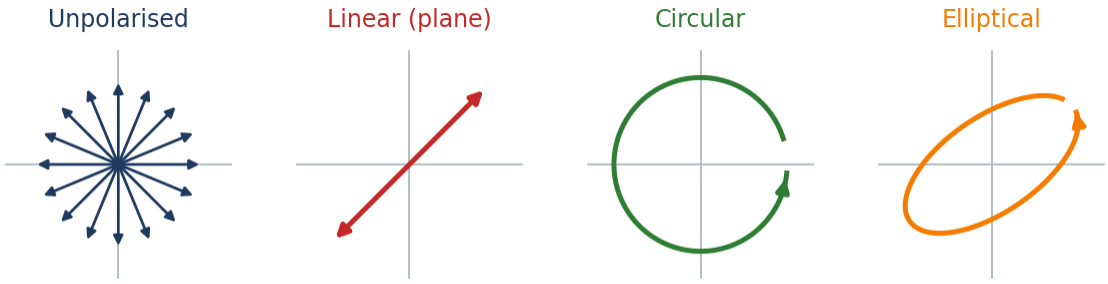
Quantity	Meaning	Expression	How to increase
Dispersive power (angular dispersion)	Angular separation dθ of two wavelengths differing by dλ	dθ/dλ = n / [(a+b) cosθ]	Higher order n, finer grating (small a+b)
Resolving power	Ability to separate two close spectral lines (Rayleigh criterion)	R = λ/dλ = nN	Higher order n, more lines N illuminated

EXAMPLE • Grating numerical

A grating has 6000 lines/cm. For λ = 5000 Å: a + b = 1/6000 cm = 1.667×10^{-6} m, so n_{max} = $1.667\times10^{-6}/5\times10^{-7}$ = 3.3, i.e. **3 orders**. For n = 2: sinθ = 2λ/(a+b) = 0.6, so θ = **36.9°**.

6 Polarisation

Light is a **transverse** electromagnetic wave. In ordinary (unpolarised) light the electric vector vibrates in all directions perpendicular to the ray. **Polarisation** restricts the vibrations to a single direction or a definite pattern. Longitudinal waves (e.g. sound) cannot be polarised.



Tip of the electric vector as seen looking against the beam (E_x–E_y plane)

Fig. 6: Types of polarisation

Type	Description	Produced by
Plane (linear)	E vibrates along one line	Polariser (Nicol, Polaroid)
Circular	Two perpendicular components, equal amplitude, phase difference π/2; tip traces a circle	Plane light at 45° through quarter-wave plate
Elliptical	Unequal amplitudes and/or phase difference other than 0, π/2, π	Plane light through a wave plate at an angle other than 45°
Partial	Mixture of polarised and unpolarised light	Reflection, scattering

Malus's law (analyser at angle θ) $$I = I_0 \cos^2\theta$$

Methods of producing plane polarised light

(i) Reflection (Brewster's law). At the particular angle of incidence i_B the reflected light is completely plane polarised (vibrations ⊥ plane of incidence) and the reflected and refracted rays are at 90°.

Brewster's law $$\tan i_B = \mu \qquad (i_B + r = 90^\circ)$$

Example: glass (μ = 1.5) → $i_B = \tan^{-1}1.5 = 56.3°$; water (μ = 1.33) → 53.1°.

(ii) Refraction (pile of plates). At i_B each glass plate reflects only the ⊥ component, so the transmitted beam gets richer in the parallel component. With about 15–20 plates the transmitted light is nearly plane polarised.

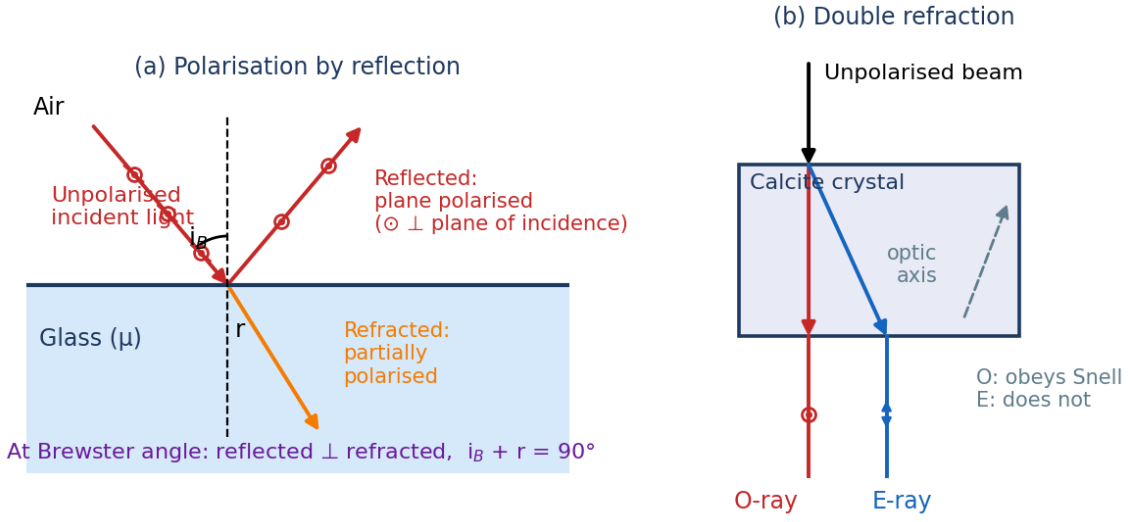


Fig. 7: (a) polarisation by reflection at the Brewster angle; (b) double refraction in calcite

7 Double Refraction and Nicol Prism

(iii) Double refraction (birefringence). A calcite or quartz crystal splits one ray into two:

- **Ordinary (O) ray:** obeys Snell's law, same speed in all directions, vibrations ⊥ principal section.
- **Extraordinary (E) ray:** does not obey Snell's law, speed varies with direction, vibrations in the principal section.
- Both are plane polarised at right angles to each other. Along the **optic axis** there is no double refraction.

- **Negative crystal** (calcite): $\mu_o > \mu_e$ (1.658 and 1.486). **Positive crystal** (quartz): $\mu_e > \mu_o$ (1.553 and 1.544).

Nicol prism

A calcite crystal (length : width = 3 : 1) with its end faces ground from 71° to 68°, cut along the diagonal and cemented with **Canada balsam** (μ = 1.55).

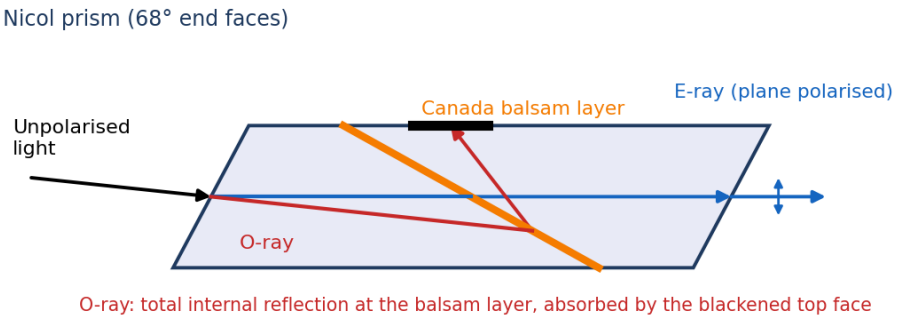


Fig. 8: Working of a Nicol prism

- Balsam is **rarer than calcite for the O-ray** (1.658 > 1.55) and denser for the E-ray (1.486 < 1.55).
- The O-ray meets the balsam at an angle greater than the critical angle (≈ 69°), so it undergoes **total internal reflection** and is absorbed by the blackened side.
- The E-ray is transmitted, so the emergent light is **plane polarised**. Used as a **polariser**, and a second Nicol acts as an **analyser**.

REMEMBER • Limitations
Limited field of view (about 29°), costly, and the emergent beam is slightly displaced laterally. Polaroid sheets are cheaper alternatives.

8 Half-Wave and Quarter-Wave Plates

A thin slice of a uniaxial crystal cut **parallel to its optic axis**. Light entering normally splits into O and E rays that travel along the same path at different speeds, so a phase difference builds up between them.

Path difference	$\Delta = (\mu_o - \mu_e)\, t$
Quarter-wave plate ($\Delta = \lambda/4$)	$t = \dfrac{\lambda}{4\,\lvert\mu_o - \mu_e\rvert}$
Half-wave plate ($\Delta = \lambda/2$)	$t = \dfrac{\lambda}{2\,\lvert\mu_o - \mu_e\rvert}$

	Quarter-wave plate (QWP)	Half-wave plate (HWP)
Phase difference	π/2 (path λ/4)	π (path λ/2)
Plane light at 45° to axis	Becomes **circularly** polarised	Stays plane, rotated by 90°
Plane light at other angle θ	Becomes elliptically polarised	Plane, rotated by 2θ
Reverse use	Circular → plane	Flips handedness of circular light



Fig. 9: Producing circularly polarised light

EXAMPLE • Brewster and Malus
A glass slab (μ = 1.5) is used at the polarising angle, so $i_B = \tan^{-1}(1.5) = 56.3°$ and r = 33.7°. Plane polarised light of intensity I_0 passes through an analyser at 60°: $I = I_0\cos^2 60° = \mathbf{I_0/4}$.

EXAMPLE • Quartz QWP
Quartz: $\mu_e = 1.553$, $\mu_o = 1.544$, $\lambda = 5890$ Å. $t = \lambda / 4(\mu_e - \mu_o) = 5.89\times10^{-7} / (4 \times 0.009) =$ **16.4 µm**.

APPLICATIONS • Applications of polarisation

- Polaroid sunglasses and camera filters cut reflected glare.
- LCD/LED displays, 3-D cinema (circular polarisers).
- Stress analysis (photoelasticity) in transparent models.
- Optical activity measurement in sugar (polarimeter); optical isolators in lasers and fibre links.

More solved problems

EXAMPLE • AR coating

Find the thickness of a MgF_2 film ($\mu = 1.38$) that makes a lens non-reflecting at $\lambda = 5500$ Å: $t = \lambda/4\mu = 5.5\times10^{-7}/(4 \times 1.38) =$ **99.6 nm**.

EXAMPLE • Resolving power

A grating with 10^4 lines illuminated fully is used in the 2nd order. $R = nN = 2 \times 10^4$. Near $\lambda = 5890$ Å, $d\lambda = \lambda/R = 5890/20000 =$ **0.29 Å**, so the sodium D lines (6 Å apart) are easily resolved.

EXAMPLE • Newton's rings in liquid

In air $D_{10}^2 - D_5^2 = 0.0145$ cm²; with a liquid it is 0.0109 cm². $\mu = 0.0145/0.0109 =$ **1.33** (water).

Key terms at a glance

Term	Meaning
Coherent sources	Sources with a constant phase difference and the same frequency
Fringe width	Distance between two consecutive bright (or dark) fringes
Optical path	Refractive index × geometrical path ($\mu \times x$)
Order (n)	Integer labelling the maxima or minima, n = 0, 1, 2 ...
Rayleigh criterion	Two lines are just resolved when the maximum of one falls on the first minimum of the other
Principal section	Plane through the optic axis and perpendicular to two opposite faces of the crystal
Analyser	Second polariser used to detect or analyse polarised light

KEY IDEA • Real-life connections

- The colours of a CD, a soap bubble and a butterfly wing are all interference or diffraction effects.
- Camera lenses look purple-green because the AR coating cancels yellow-green reflections best.
- Polaroid sunglasses block horizontally polarised glare reflected from roads and water.

9 Formula Sheet and Exam Corner

Topic	Key formulae
Thin film (reflected)	$2\mu t \cos r = (2n+1)\lambda/2$ (bright); $= n\lambda$ (dark). AR coating: $t = \lambda/4\mu$
Newton's rings	$D_n^2 = 4n\lambda R$ (dark); $\lambda = (D_{n+p}^2 - D_n^2)/4pR$; μ = slope(air)/slope(liquid)
Single slit	$a \sin\theta = n\lambda$ (minima); $I = I_0 (\sin\alpha/\alpha)^2$
Grating	$(a+b) \sin\theta = n\lambda$; $d\theta/d\lambda = n/[(a+b)\cos\theta]$; $R = nN$
Polarisation	Malus: $I = I_0\cos^2\theta$; Brewster: $\tan i_B = \mu$
Wave plates	QWP: $t = \lambda/4\lvert\mu_o - \mu_e\rvert$; HWP: $t = \lambda/2\lvert\mu_o - \mu_e\rvert$

EXAM CORNER • Short questions (2–3 marks)

- State the principle of superposition and the conditions for sustained interference.
- Why is the centre of Newton's rings dark in reflected light?
- Differentiate between Fresnel and Fraunhofer diffraction.
- What is a diffraction grating? Write the grating equation.
- State Brewster's law and Malus's law.
- Define the O-ray, E-ray and optic axis.
- Why do thin films show colours in white light?

EXAM CORNER • Long questions (10 marks)

- Derive the conditions for bright and dark fringes in a thin film in reflected light. Explain colours in thin films.
- Explain the formation of Newton’s rings. Derive the expression for the diameter of dark rings and describe how the wavelength is determined.
- Describe Fraunhofer diffraction at a single slit and obtain the intensity distribution with a neat diagram.
- Explain the construction and working of a Nicol prism.
- Explain half-wave and quarter-wave plates, with the thickness expressions.
- Write notes on the dispersive power and resolving power of a grating.

EXAM CORNER • Quick MCQs (answers at the end of the box)

- 1. The centre of Newton’s rings in reflected light is (a) bright (b) dark (c) coloured.
- 2. At the Brewster angle the angle between reflected and refracted rays is (a) 0° (b) 45° (c) 90°.
- 3. Resolving power of a grating with N lines in order n is (a) n/N (b) nN (c) N/n.
- 4. A quarter-wave plate turns 45° plane polarised light into (a) circular (b) unpolarised (c) rotated plane light.
- **Answers:** 1-b, 2-c, 3-b, 4-a.

QUICK SUMMARY

- Interference = redistribution of energy; needs coherent sources. Thin film: $\Delta = 2\mu t \cos r$ with an extra $\lambda/2$ in reflection.
- Diffraction = bending around apertures; grating gives sharp orders with high dispersive and resolving power.
- Polarisation proves light is transverse; it is produced by reflection, refraction, double refraction and wave plates.

UNIT

II

ENGINEERING PHYSICS • DIGITAL NOTES

Crystallography and X-ray Diffraction

Space lattice • Basis • Unit cell • Bravais lattices • Packing fraction • Miller indices • Bragg's law • Diffractometer • Laue and powder methods

QUICK SUMMARY • What you will learn

- How atoms are arranged in crystals and how to describe that arrangement in a language of lattices and cells
- How to label planes with Miller indices and calculate the distance between them
- How X-rays reveal the inside of a crystal (Bragg's law, Laue and powder methods)



Fig. 1: Roadmap of Unit II

1 Space Lattice, Basis and Unit Cell

Crystalline solids	Amorphous solids
Long-range, periodic order of atoms	No long-range order (short-range only)
Sharp melting point; anisotropic	Softens over a range; isotropic
Examples: NaCl, Cu, diamond, quartz	Examples: glass, rubber, plastics

Space lattice: an infinite array of points in space such that every point has **identical surroundings**.

Basis: the atom or group of atoms attached to every lattice point.

Crystal structure = Lattice + Basis.

Unit cell: the smallest volume (parallelepiped) which, repeated in three dimensions by translation, builds the whole crystal.

Primitive cell: a unit cell with lattice points only at its corners (one lattice point per cell). Non-primitive cells have extra points at the body or face centres.

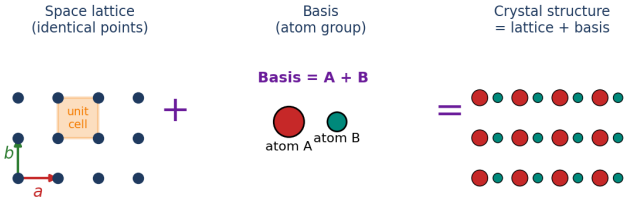


Fig. 2: Lattice, basis and crystal structure (2-D view)

- Edges a, b, c and angles α (between b, c), β (between c, a) and γ (between a, b) are the six **lattice parameters**.
- Translation vector: $\mathbf{T = n_1 a + n_2 b + n_3 c}$ (n_1, n_2, n_3 integers) takes any lattice point to an equivalent one.

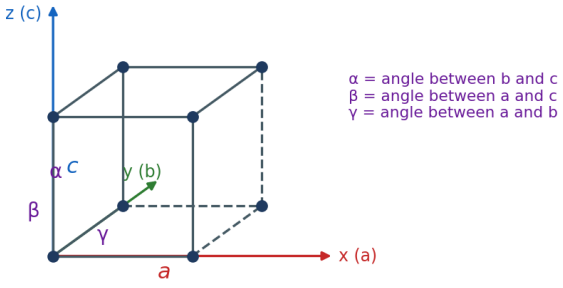


Fig. 3: Unit cell and lattice parameters

2 Crystal Systems and Bravais Lattices

On the basis of the relations among a, b, c and α, β, γ, all crystals fall into **7 crystal systems**. Adding centring (P = primitive, I = body centred, F = face centred, C = base centred) gives exactly **14 Bravais lattices** (Bravais, 1848).

System	Axes	Angles	Bravais lattices	Examples
Cubic	a = b = c	α = β = γ = 90°	P, I, F (3)	Cu, NaCl, diamond
Tetragonal	a = b ≠ c	α = β = γ = 90°	P, I (2)	White tin, TiO_2
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	P, C, I, F (4)	$BaSO_4$, rhombic sulphur
Monoclinic	a ≠ b ≠ c	α = γ = 90° ≠ β	P, C (2)	Gypsum, monoclinic S
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ ≠ 90°	P (1)	$CuSO_4 \cdot 5H_2O$
Trigonal (rhombohedral)	a = b = c	α = β = γ ≠ 90°	P (1)	Calcite, Bi
Hexagonal	a = b ≠ c	α = β = 90°, γ = 120°	P (1)	Zn, Mg, graphite

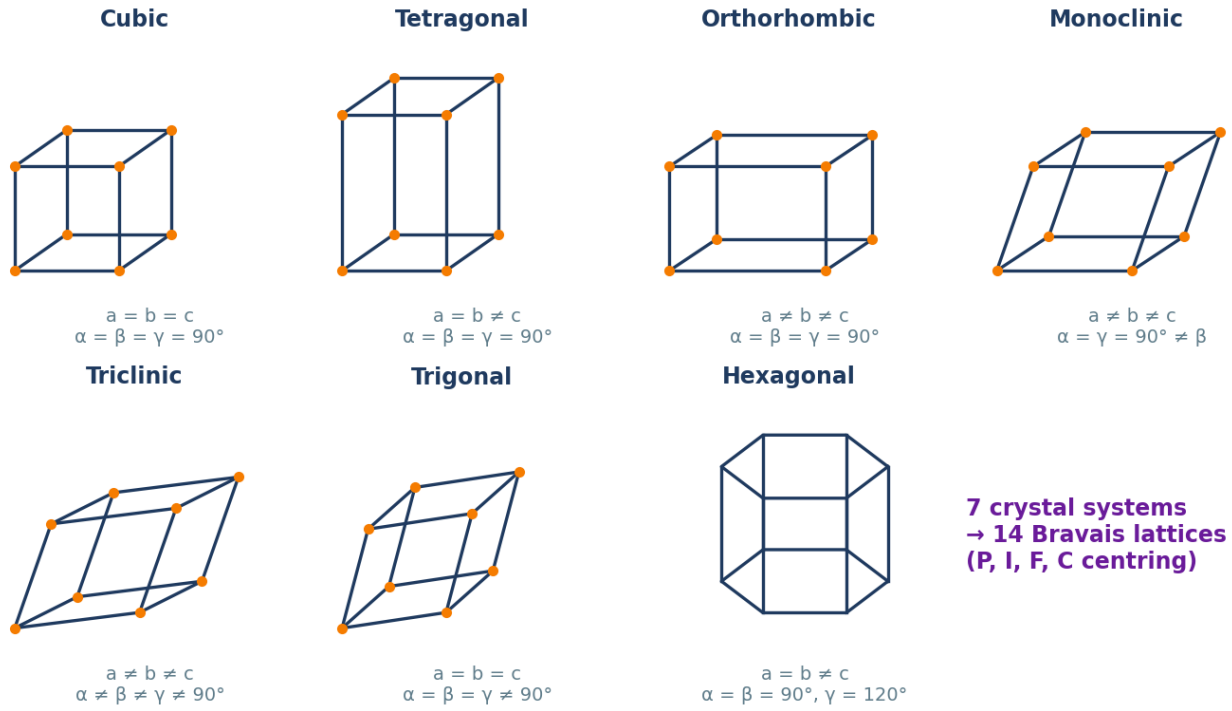


Fig. 4: Unit cells of the seven crystal systems

REMEMBER • Counting trick

3 + 2 + 4 + 2 + 1 + 1 + 1 = **14**. Only the cubic system has all three of P, I and F; only orthorhombic has all four types.

3 Cubic Structures: Coordination Number and Packing Fraction

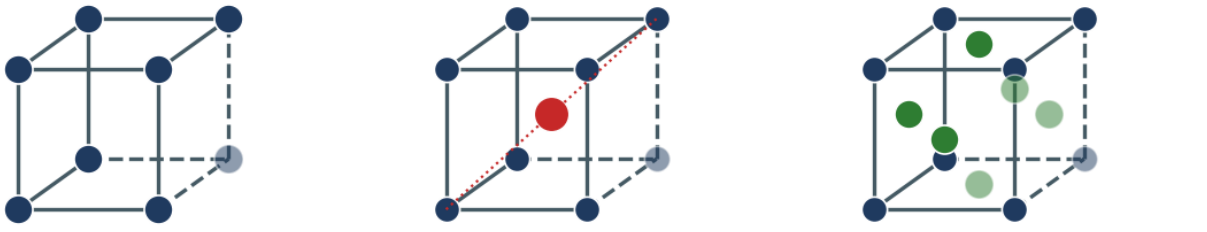


Fig. 5: The three cubic Bravais lattices and the number of atoms they contain

DEFINITION • Coordination number (N)

The number of nearest neighbours of an atom in the crystal.

DEFINITION • Atomic packing fraction (APF)

APF = (volume of atoms in a unit cell) / (volume of the unit cell) = $n \cdot (4/3)\pi r^3 / a^3$, where n = atoms per cell.

Property	SC	BCC	FCC
Atoms per cell (n)	1	2	4
Coordination number	6	8	12
Touching atoms along	Cube edge	Body diagonal	Face diagonal
Relation of r and a	a = 2r	a = 4r/√3	a = 4r/√2 = 2√2 r
Nearest-neighbour distance	a	√3 a/2	a/√2
Packing fraction	π/6 = 0.52 (52%)	√3π/8 = 0.68 (68%)	√2π/6 = 0.74 (74%)
Examples	Polonium	Fe, Na, Cr	Cu, Al, Ag, Au

SC (r = a/2, n = 1)

$$APF = \frac{1 \times \frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3} = \frac{\pi}{6} = 0.52$$

BCC (r = √3a/4, n = 2)

$$APF = \frac{2 \times \frac{4}{3}\pi\left(\frac{\sqrt{3}a}{4}\right)^3}{a^3} = \frac{\sqrt{3}\pi}{8} = 0.68$$

FCC (r = a/2√2, n = 4)

$$APF = \frac{4 \times \frac{4}{3}\pi\left(\frac{a}{2\sqrt{2}}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$

EXAMPLE • Copper

Cu is FCC with r = 1.278 Å. a = 2√2 r = **3.615 Å**. Density $\rho = nM/(N_A a^3) = 4 \times 63.5\times10^{-3} / (6.022\times10^{23} \times (3.615\times10^{-10})^3) \approx$ **8.9 × 10³ kg/m³**.

Beyond the simple cubic family

Structure	Atoms / cell	Coordination no.	Packing fraction	Examples
HCP (c/a = 1.633)	6 (2 per primitive cell)	12	0.74	Zn, Mg, Ti
Diamond cubic	8	4	0.34	C (diamond), Si, Ge
NaCl (rock salt)	4 Na^+ + 4 Cl^-	6	—	NaCl, KCl, MgO

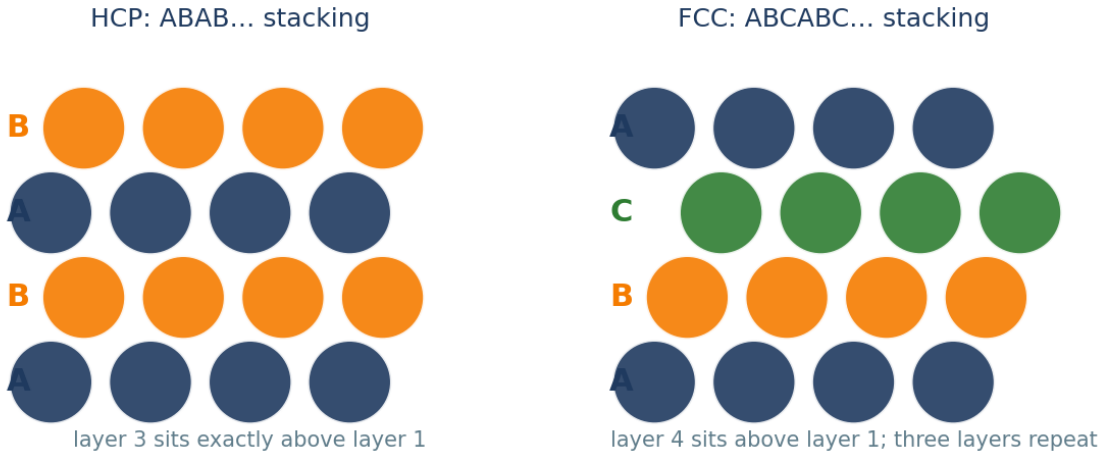


Fig. 6: Close-packed layers: HCP repeats every two layers, FCC every three

KEY IDEA • Remember

FCC (and HCP) are **close-packed** structures: 74% of the space is filled and 26% is empty. Tight packing is why many metals (Cu, Al, Au) are ductile.

4 Miller Indices

DEFINITION • Miller indices (h k l)

A set of three smallest integers, proportional to the reciprocals of the intercepts a plane makes on the crystallographic axes, used to label planes in a crystal.



Fig. 7: Procedure for finding Miller indices

EXAMPLE • Worked example

A plane cuts intercepts 2a, 3b, 6c. Intercepts (2, 3, 6) → reciprocals (1/2, 1/3, 1/6) → multiply by LCM 6 → **(3 2 1)**.
A plane parallel to the z-axis has intercept ∞ on z, so l = 1/∞ = 0, e.g. intercepts (1, 1, ∞) → **(1 1 0)**.

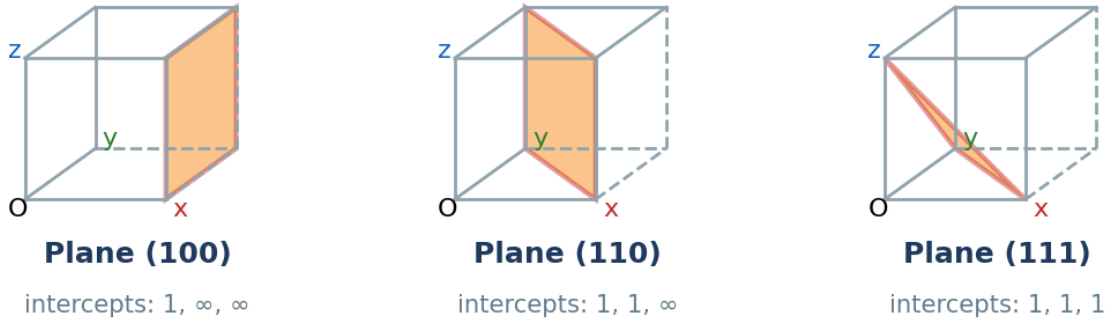


Fig. 8: The (100), (110) and (111) planes of a cubic cell

- **Parallel planes** have the same Miller indices.
- A negative intercept is written with a bar on top, e.g. ($1\bar{1}$ 0) is read as “one-bar one zero”.
- (h k l) = one plane; {h k l} = family of equivalent planes; [u v w] = a direction; ⟨u v w⟩ = family of directions.
- Larger indices mean closer spaced planes and lower atomic density per plane.

Separation between successive (hkl) planes

Let the plane nearest to the origin make intercepts a/h, b/k, c/l (points A, B, C) and let ON = d be the perpendicular from the origin. If the normal makes angles α, β, γ with the axes, then cosα = d/(a/h) = hd/a, cosβ = kd/b, cosγ = ld/c. For orthogonal axes $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$, which gives the results below.

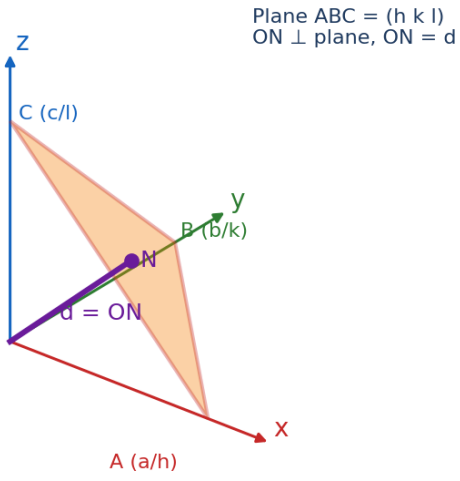


Fig. 9: Geometry for d(hkl)

General (orthogonal axes)	$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$
Cubic crystal (a = b = c)	$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$
Ratio of spacings in cubic	$d_{100} : d_{110} : d_{111} = 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$

EXAMPLE • Numerical

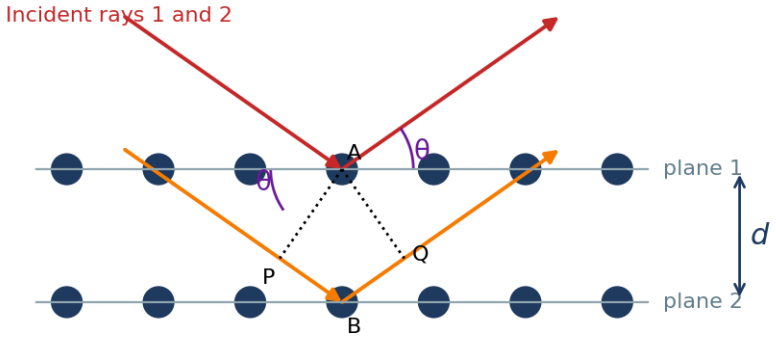
For Cu (a = 3.615 Å): d_{111} = 3.615/√3 = **2.087 Å**; d_{200} = 3.615/2 = **1.808 Å**; d_{220} = 3.615/√8 = **1.278 Å**.

5 X-ray Diffraction and Bragg's Law

DEFINITION • X-rays

Electromagnetic waves of wavelength 0.1–10 Å, produced when fast electrons strike a metal target. The spectrum has a continuous part (shortest wavelength λ_{min} = 12400/V Å, with V in volts) and sharp characteristic lines (Kα, Kβ) of the target element.

X-rays have wavelengths of about **0.1–10 Å**, comparable to interatomic spacings in crystals, so a crystal behaves like a natural three-dimensional diffraction grating for X-rays. Bragg (1913) treated diffraction as "reflection" from successive atomic planes.



Path difference = PB + BQ = 2d sinθ → 2d sinθ = nλ

Fig. 10: Reflection of X-rays from two successive planes of spacing d

Rays 1 and 2 reflect from planes a distance d apart at glancing angle θ. The extra path travelled by ray 2 is PB + BQ = d sinθ + d sinθ. For constructive interference this must be a whole number of wavelengths.

Bragg's law	$2d\sin\theta = n\lambda \qquad n = 1, 2, 3, \ldots$
Cubic crystal (n = 1)	$\sin^2\theta = \frac{\lambda^2}{4a^2}(h^2 + k^2 + l^2)$
Maximum order possible	$n_{max} = \frac{2d}{\lambda} \quad (\sin\theta \le 1 \Rightarrow \lambda \le 2d)$

REMEMBER • Points to note

- Bragg's law works only if λ ≤ 2d: visible light cannot be diffracted by atomic planes.
- θ is the **glancing angle** (angle with the plane), not with the normal; the diffracted beam is deviated by 2θ from the incident beam.
- Unlike ordinary reflection, Bragg reflection occurs only at specific angles.

EXAMPLE • Numerical

NaCl planes have d = 2.82 Å. For X-rays of λ = 1.54 Å: first order sinθ = 1.54/(2 × 2.82) = 0.273, so θ = **15.8°**. Second order: sinθ = 0.546, θ = **33.1°**.

6 X-ray Diffractometer

A diffractometer records the intensity of diffracted X-rays electronically against the angle 2θ, usually for a powder sample. It replaces the photographic film with a counter, so data are fast, accurate and digital.

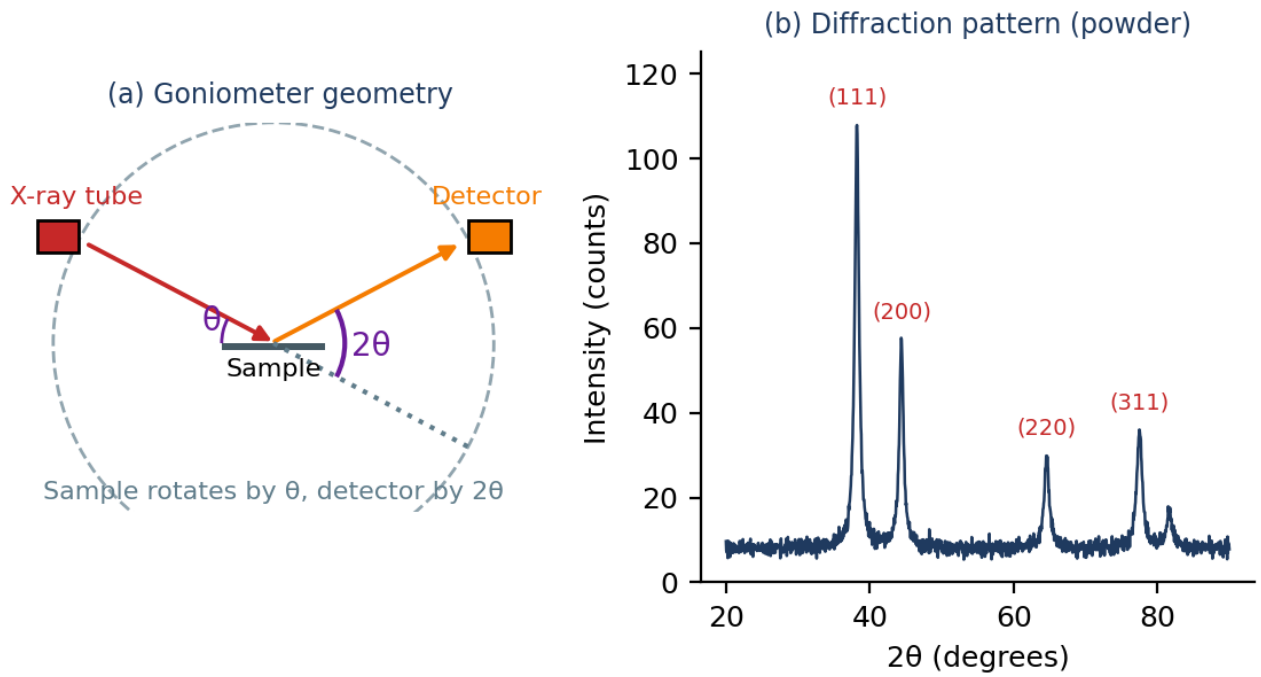


Fig. 11: Goniometer geometry and a typical powder pattern (peak positions give d, peak heights give atomic arrangement)

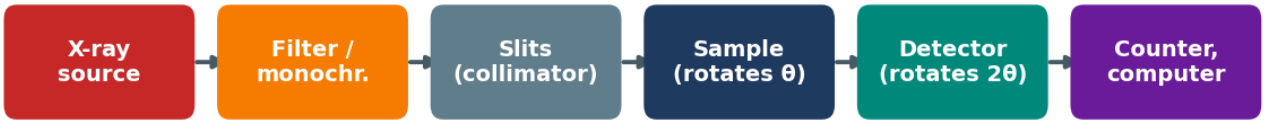


Fig. 12: Block diagram of an X-ray diffractometer

Part	Function
X-ray tube	Fast electrons strike a metal target (Cu, λ(Kα) = 1.54 Å; Mo) to produce characteristic X-rays
Filter / monochromator	Ni foil removes Kβ radiation so that the beam is almost monochromatic
Slits (collimator)	Make a narrow, nearly parallel beam; also limit scattering
Sample stage (goniometer)	Holds the powder flat; rotates through θ at constant angular speed
Detector	Scintillation or proportional counter rotating at 2θ collects the diffracted beam
Ratemeter + computer	Convert pulses to counts; plot intensity versus 2θ

- **Output:** peaks at angles that satisfy Bragg's law; compare with standard database (JCPDS/ICDD) to identify phases.
- **Uses:** phase identification, lattice parameters, crystallite size (Scherrer: $D = K\lambda / \beta\cos\theta$), strain, percentage crystallinity.

7 Crystal Structure Determination: Laue and Powder Methods

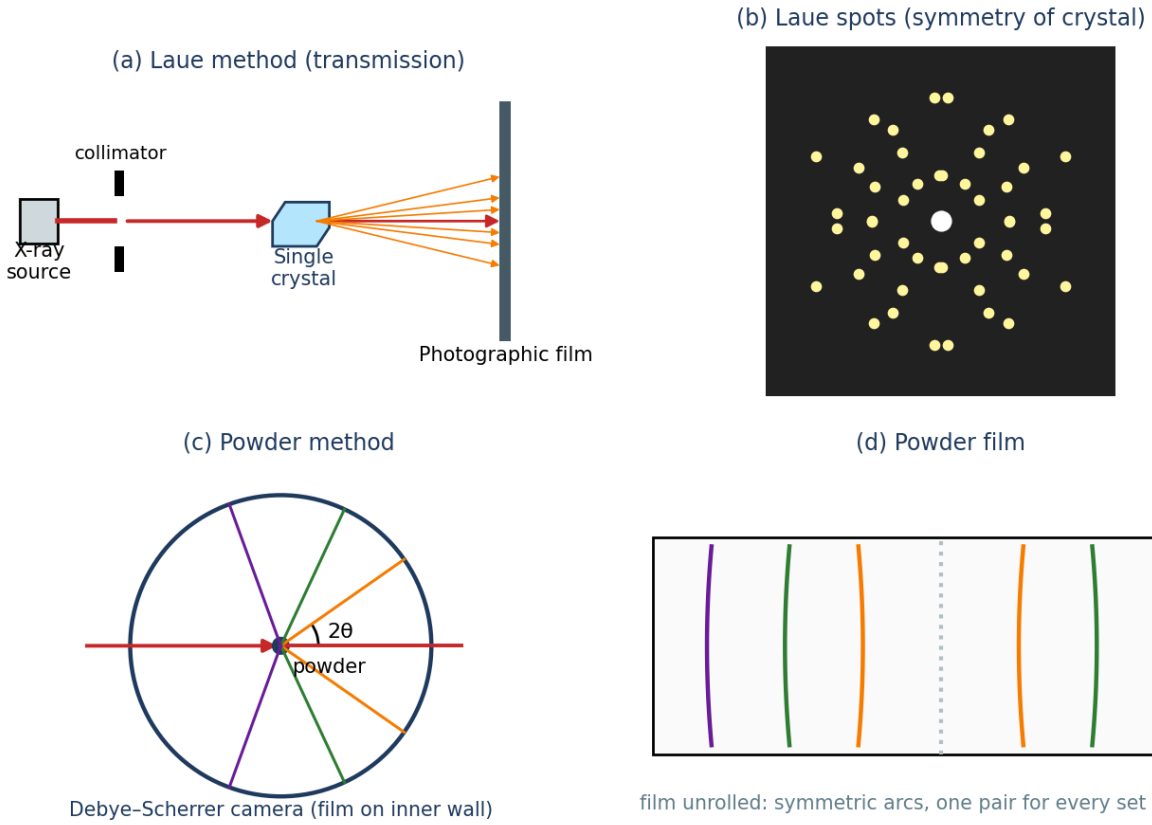


Fig. 13: Laue method (a, b) and Debye–Scherrer powder method (c, d)

Feature	Laue method	Powder (Debye-Scherrer) method
Sample	Single crystal, fixed	Fine powder or polycrystal (random orientation)
X-rays	**White** (continuous λ)	**Monochromatic**
Idea	For each plane set, the crystal picks the λ that satisfies 2d sinθ = nλ	Among many crystallites, some are always oriented at the Bragg angle for each (hkl)
Pattern	Spots (symmetric dot pattern)	Concentric cones → arcs on film
Gives	Crystal symmetry and orientation; checks perfection	Lattice type, d-spacings, lattice constant
Variants	Transmission and back-reflection	Film camera or diffractometer

Powder method: step-by-step

- Measure the distance S between a symmetric pair of arcs on the film. For camera radius R: $4\theta = S/R$ (in radians), so $\theta = S/4R$.
- Find d from Bragg's law, $d = \lambda/(2\sin\theta)$, for each line.
- Compute $\sin^2\theta$ for each line and compare the ratios with the table to find the lattice type.
- Index the lines (h k l) and compute the lattice constant $a = \lambda\sqrt{(h^2+k^2+l^2)} / (2\sin\theta)$.

Cubic lattice	Allowed reflections	$\sin^2\theta$ ratio ($\propto h^2+k^2+l^2$)
Simple cubic	all (hkl)	1 : 2 : 3 : 4 : 5 : 6 : 8 : 9 ... (7 missing)
BCC	h + k + l even	2 : 4 : 6 : 8 : 10 : 12 ... = 1 : 2 : 3 : 4 : 5 : 6 ...
FCC	h, k, l all odd or all even	3 : 4 : 8 : 11 : 12 : 16 ...

KEY IDEA • Quick identification
Take the ratio of the first two lines: **0.75 (3:4)** suggests FCC; **1:2 pattern** with the first line = 2 suggests BCC; the series 1:2:3:4:5:6 with a missing 7 suggests SC.

APPLICATIONS • Applications of X-ray diffraction

- Identifying unknown materials and phases by their “fingerprint” pattern.
- Finding lattice constants, crystal size, strain and defects in metals and nanomaterials.
- Determining structure of proteins and DNA (X-ray crystallography); quality control in pharmaceuticals and semiconductors.

8 Worked Problems

EXAMPLE • 1. Packing

BCC iron has a = 2.87 Å. r = √3a/4 = **1.24 Å**; nearest-neighbour distance = 2r = 2.48 Å; density = 2 × $55.85\times10^{-3}/(6.022\times10^{23} \times (2.87\times10^{-10})^3)$ = **7.85×10^3 kg/m³**.

EXAMPLE • 2. Miller indices

A plane has intercepts a, b/2, c/3 → reciprocals 1, 2, 3 → **(1 2 3)**. Its spacing in a cubic crystal with a = 4.2 Å: d = 4.2/√14 = **1.12 Å**.

EXAMPLE • 3. Bragg’s law

For d = 1.1 Å, the maximum order for λ = 0.5 Å: n_{max} = 2d/λ = 4.4 → **4**. First-order glancing angle: sinθ = 0.5/2.2 = 0.227, θ = **13.1°**.

EXAMPLE • 4. Powder pattern

A cubic powder (λ = 1.54 Å) shows sin²θ = 0.1365, 0.1820, 0.3640, 0.5005. Dividing by 0.0455 gives **3 : 4 : 8 : 11** → **FCC** with lines (111), (200), (220), (311). a = λ√3/(2 sinθ) = 1.54 × 1.732 / (2 × 0.3695) = **3.61 Å** (copper).

EXAM CORNER • Try yourself (answers in brackets)

- Aluminium is FCC with a = 4.05 Å. Find the atomic radius and density. [1.43 Å; 2.7 × 10³ kg/m³]
- Find the Miller indices of a plane with intercepts 3a, 2b, c. [(2 3 6)]
- Silicon has a = 5.43 Å. Find d_{220}. [1.92 Å]
- X-rays of λ = 1.54 Å are reflected from planes with d = 2.5 Å. Find the first-order Bragg angle. [17.9°]
- The first two powder lines of a cubic crystal have sin²θ in the ratio 1 : 2. Which lattice is this likely to be, and what are the indices of the lines? [BCC; (110), (200)]

Key terms at a glance

Term	Meaning
Lattice point	A point with identical surroundings to every other point of the lattice
Primitive vectors	Smallest non-coplanar translation vectors a, b, c of the lattice
Glancing angle	Angle θ between the incident X-ray beam and the atomic plane
Systematic absence	A reflection missing because of centring (e.g. BCC needs h+k+l even)
Anisotropy	Dependence of physical properties on crystal direction

Diffraction by X-rays, electrons and neutrons

	X-rays	Electrons	Neutrons
Interacts with	Electron cloud	Electrostatic potential (strong)	Nuclei (and magnetic moments)
Penetration	Moderate (μm–mm)	Very small (surface, thin films)	Large (bulk)
Wavelength from	Target metal / tube voltage	λ = h/√(2meV)	λ = h/√(2m_nE)
Best for	General structure	Surfaces, thin films, TEM	Light atoms (H), magnetic structure

9 Formula Sheet and Exam Corner

Topic	Key formulae
Lattice	Crystal = lattice + basis; 7 systems, 14 Bravais lattices
Atoms / cell, N	SC: 1, 6; BCC: 2, 8; FCC: 4, 12
APF	SC 0.52, BCC 0.68, FCC 0.74. Density $\rho = nM / (N_A a^3)$
Miller indices	Reciprocals of intercepts → smallest integers (h k l)
Plane spacing	$d = a/\sqrt{(h^2+k^2+l^2)}$ (cubic); $1/d^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$
Bragg	$2d \sin\theta = n\lambda$; $a = \lambda\sqrt{(h^2+k^2+l^2)}/(2 \sin\theta)$
Powder film	$\theta = S / 4R$ (radians)

EXAM CORNER • Short questions (2–3 marks)

- Define space lattice, basis and unit cell.
- List the seven crystal systems with their axial relations.
- Define coordination number and packing fraction.
- Calculate the packing fraction of an FCC structure.
- What are Miller indices? Find the Miller indices of a plane with intercepts 2a, 3b, ∞.
- State Bragg’s law. Why can visible light not be used?

EXAM CORNER • Long questions (10 marks)

- Describe the 14 Bravais lattices and 7 crystal systems.
- Derive the packing fractions of SC, BCC and FCC.
- Derive an expression for the interplanar spacing in terms of Miller indices for a cubic lattice.
- Derive Bragg’s law. Explain the working of an X-ray diffractometer with a neat sketch.
- Explain Laue’s method and the powder method for crystal structure determination.

EXAM CORNER • Quick MCQs

- 1. Number of atoms in an FCC unit cell: (a) 2 (b) 4 (c) 6.
- 2. Which has the highest packing fraction? (a) SC (b) BCC (c) FCC.
- 3. d_{110} in a cubic crystal of edge a = (a) a (b) $a/\sqrt{2}$ (c) $a/\sqrt{3}$.
- 4. The Laue method uses (a) monochromatic X-rays (b) white X-rays.
- **Answers:** 1-b, 2-c, 3-b, 4-b.

QUICK SUMMARY

- A crystal is a lattice with a basis; 14 Bravais lattices belong to 7 systems.
- Packing: SC 52%, BCC 68%, FCC 74%.
- $d_{hkl} = a/\sqrt{(h^2+k^2+l^2)}$; Bragg: $2d \sin\theta = n\lambda$. Laue uses white X-rays on a single crystal, the powder method uses monochromatic X-rays on powder.

UNIT

III

ENGINEERING PHYSICS • DIGITAL NOTES

Dielectric and Magnetic Materials

Polarisation • Electric vectors • Electronic, ionic and orientation polarisation • Lorentz field • Clausius–Mossotti • Dielectric loss • Dia, para, ferro, antiferro, ferri magnetism • Domains • Hysteresis

QUICK SUMMARY • What you will learn

- How an insulator stores energy in an electric field and why its dielectric constant depends on frequency
- How the atomic magnetic moment leads to five classes of magnetic materials
- Why ferromagnets form domains, show hysteresis and are used as soft or hard magnets

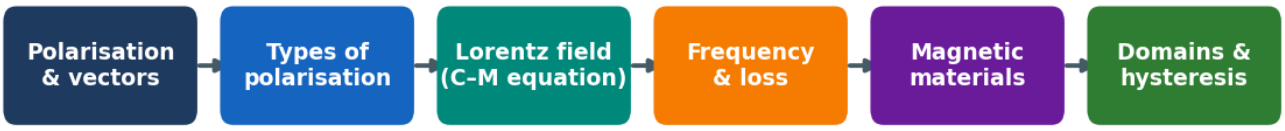


Fig. 1: Roadmap of Unit III

1 Dielectric Materials and Polarisation

DEFINITION • Dielectric

An insulating material with no free charges that can be **polarised** by an electric field. The field only shifts bound charges slightly, so no current flows. Examples: glass, mica, ceramics, polymers, water.

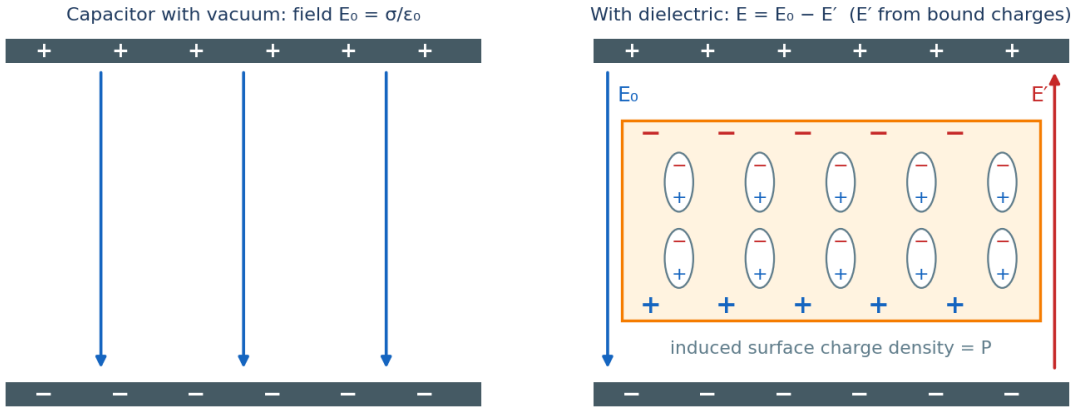


Fig. 2: A dielectric slab between capacitor plates: bound surface charges reduce the field inside

- **Dipole moment:** two equal and opposite charges ±q separated by a distance d form a dipole, μ = q d (C·m).
- **Dielectric polarisation (P):** dipole moment per unit volume, P = Nμ (C/m²). It equals the induced surface charge density.
- **Polar** molecules (H_2O, HCl) have permanent dipoles; **non-polar** molecules (H_2, N_2, CH_4) get induced dipoles only under a field.

Quantity	Symbol and definition	Unit
Polarisability	α = μ / E: induced dipole moment per unit field	F·m²
Polarisation	P = Nμ = NαE: dipole moment per unit volume	C/m²
Electric susceptibility	$\chi_e = P / (\varepsilon_0 E)$	dimensionless
Dielectric constant	$\varepsilon_r = \varepsilon/\varepsilon_0 = C_{medium}/C_{vacuum}$	dimensionless
Displacement vector	D = ε₀E + P: depends on free charge only (equals σ of plates)	C/m²

Relation between the electric vectors

Displacement vector	$D = \varepsilon_0 E + P = \varepsilon_0 \varepsilon_r E$
Polarisation	$P = \varepsilon_0 \chi_e E = \varepsilon_0 (\varepsilon_r - 1) E$
Susceptibility and ε_r	$\varepsilon_r = 1 + \chi_e$
Polarisation from atoms	$P = N\alpha E_i \quad (E_i = \text{local field})$

EXAMPLE • Numerical
A dielectric with $\varepsilon_r = 4$ is in a field $E = 10^5$ V/m. $\chi_e = 3$; $P = \varepsilon_0 \chi_e E = 8.85\times10^{-12} \times 3 \times 10^5 =$ **2.66×10^{-6} C/m²**; $D = \varepsilon_0 \varepsilon_r E =$ **3.54×10^{-6} C/m²**.

2 Types of Polarisation

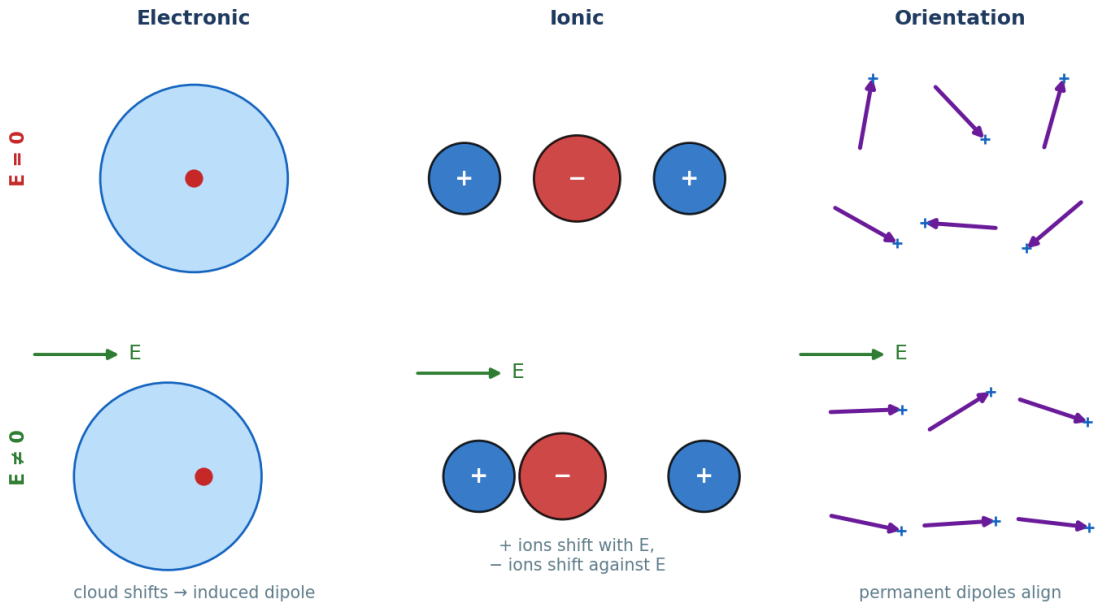


Fig. 3: Electronic, ionic and orientation polarisation without and with an applied field

Type	Cause	Present in	Time to set up	Temperature?
Electronic	Electron cloud shifts relative to nucleus	All materials	≈10^{-15} s (optical)	Independent
Ionic	+ and − ions displace in opposite directions	Ionic solids (NaCl, KCl)	≈10^{-12}–10^{-13} s (IR)	Almost independent
Orientation (dipolar)	Permanent dipoles line up with the field	Polar molecules (H_2O, HCl)	≈10^{-10} s or longer (micro/radio)	∝ 1/T
Space charge (interfacial)	Charges pile up at grain boundaries and interfaces	Impure / multiphase solids	Slow (audio and below)	Depends

Total polarisability	$\alpha = \alpha_e + \alpha_i + \alpha_o$

(a) Electronic polarisability (quantitative)

Model an atom as a nucleus +Ze at the centre of a uniform spherical electron cloud of radius R (total charge −Ze). Under field E the nucleus is displaced by x from the cloud centre. Only the charge inside radius x pulls back, so the restoring force is $F_r = Ze \times [Ze\, x / (4\pi\varepsilon_0 R^3)]$. Equilibrium: $F_r = ZeE$.

Displacement	$x = \frac{4\pi\varepsilon_0 R^3 E}{Ze}$
Induced dipole $\mu = Zex$	$\mu_e = 4\pi\varepsilon_0 R^3 E \Rightarrow \alpha_e = 4\pi\varepsilon_0 R^3$
Polarisation	$P_e = N\alpha_e E$

So α_e grows as R^3, and it is independent of temperature. Larger atoms are easier to polarise.

(b) Ionic polarisability (quantitative)

In an ionic pair (e.g. Na^+ Cl^-) the field shifts the positive ion by x_1 along E and the negative ion by x_2 against E. If β_1 and β_2 are the restoring-force constants, equilibrium gives $\beta_1 x_1 = eE$ and $\beta_2 x_2 = eE$.

Dipole moment $\mu = e(x_1 + x_2)$	$\mu_i = e^2 E\left(\frac{1}{\beta_1} + \frac{1}{\beta_2}\right)$
Ionic polarisability	$\alpha_i = e^2\left(\frac{1}{\beta_1} + \frac{1}{\beta_2}\right)$
With $\beta = m\omega_0^2$ (reduced mass m)	$\alpha_i = \frac{e^2}{\omega_0^2}\left(\frac{1}{m} + \frac{1}{M}\right)$

(c) Orientation polarisation (qualitative)

Polar molecules have permanent dipoles μ, normally randomised by thermal motion. A field gives each dipole a torque $\mu E \sin\theta$ that tries to align it, thermal agitation resists. For ordinary fields ($\mu E \ll kT$) the Langevin theory gives:

Orientation polarisability	$\alpha_o = \frac{\mu^2}{3kT} \qquad P_o = \frac{N\mu^2 E}{3kT}$

KEY IDEA • Temperature test

Only orientation polarisation falls as 1/T. So for polar dielectrics ε_r decreases when heated, while for non-polar dielectrics ε_r is nearly constant with temperature.

3 Internal (Lorentz) Field and Clausius–Mossotti Equation

An atom inside a solid feels the applied field **plus** the fields of all neighbouring dipoles. This **local (internal) field** E_i is larger than the macroscopic field E. Lorentz imagined a small sphere (cavity) around the atom: dipoles outside are treated as a continuum, those inside are summed individually.

Contributions to the field at the centre of the cavity:

- **E**: macroscopic field in the dielectric (applied field and the field of charges on the outer faces).
- **$E_2 = P/3\varepsilon_0$**: field due to polarisation charges on the cavity wall.
- **$E_3 = 0$**: field of the dipoles inside the cavity, which is zero for cubic symmetry or random arrangement.

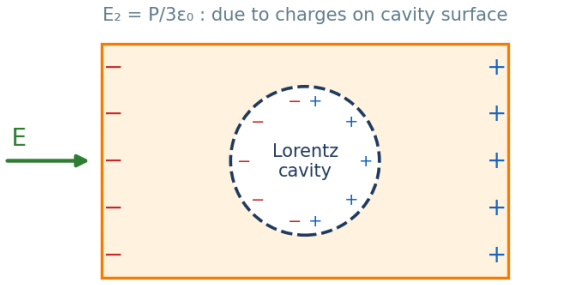


Fig. 4: Lorentz cavity in a polarised dielectric

Lorentz internal field	$E_i = E + \frac{P}{3\varepsilon_0}$

Substitute in $P = N\alpha E_i$ and use $P = \varepsilon_0(\varepsilon_r - 1)E$:

Clausius–Mossotti equation	$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0}$
Using $N = N_a\rho/M$	$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} \cdot \frac{M}{\rho} = \frac{N_A\alpha}{3\varepsilon_0}$

REMEMBER • Importance and limits

- It links a **macroscopic** property (ε_r) to a **microscopic** one (α), so α can be found from measured ε_r and density.
- Valid for elemental solids, non-polar gases and cubic crystals. Fails for strongly polar liquids such as water.

EXAMPLE • Numerical
A gas has N = 2.7 × 10^{25} m^{-3} and α = 2.2 × 10^{-40} F·m². Nα/3ε₀ = 2.24 × 10^{-4} = (ε_r−1)/(ε_r+2) → ε_r ≈ **1.00067**.

4 Complex Dielectric Constant, Frequency Dependence and Loss

Under an AC field the polarisation lags behind the field because dipoles need time to respond. The dielectric constant is therefore treated as a **complex** quantity.

Complex dielectric constant	$\varepsilon^* = \varepsilon' - j\varepsilon''$
Loss tangent	$\tan\delta = \frac{\varepsilon''}{\varepsilon'}$
Debye relaxation (τ = relaxation time)	$\varepsilon' = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1+\omega^2\tau^2}, \quad \varepsilon'' = \frac{(\varepsilon_s - \varepsilon_\infty)\,\omega\tau}{1+\omega^2\tau^2}$
Power loss per unit volume	$P_{loss} = \omega\,\varepsilon_0\,\varepsilon'\tan\delta\,E_{rms}^2$

- ε′ (real part) = energy **stored**; ε″ (imaginary part) = energy **dissipated as heat** (dielectric loss).
- δ = loss angle, the angle by which the current in a real capacitor falls short of 90° ahead of voltage.

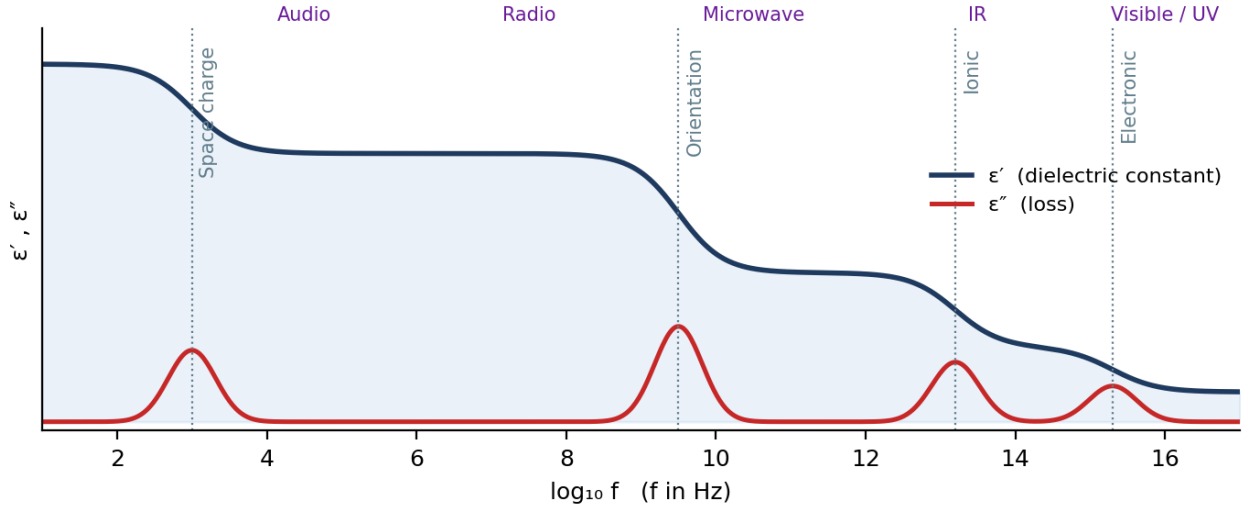


Fig. 5: Variation of ε′ and ε″ with frequency: each mechanism drops out when f exceeds its natural response rate, giving a step in ε′ and a peak in loss

- **Low frequency:** all four mechanisms follow the field, ε′ is largest.
- Above ~10^3 Hz space-charge polarisation can no longer follow; above ~10^9–10^{10} Hz (microwave) orientation polarisation drops out; above ~10^{13} Hz (IR) ionic drops out; only electronic remains up to ~10^{15} Hz (UV).
- At optical frequency $\varepsilon_r = n^2$ (refractive index squared) because only electronic polarisation survives.

APPLICATIONS • Dielectric loss matters in
- Capacitors: low tanδ is needed to avoid heating (mica, polystyrene).
- Microwave ovens: water's loss peak is used to heat food.
- Power cables and transformer oils: loss limits efficiency and causes breakdown.

5 Magnetic Materials: Basic Quantities

A magnetic dipole is a pair of poles ±p separated by l, or equivalently a current loop. Every magnetic property of matter arises from such moving charges.

Moment of a current loop	$m = IA$
Magnetisation (moment per volume)	$M = \frac{\sum m}{V}$
Flux density	$B = \mu_0(H + M) = \mu_0\mu_r H$
Susceptibility	$\chi = \frac{M}{H}, \qquad \mu_r = 1 + \chi$

Quantity	Meaning	Unit
Magnetic moment m	Strength of a magnetic dipole	A·m²
Magnetisation M	Magnetic moment per unit volume	A/m
Magnetic field H	Field due to external currents	A/m
Flux density B	Total field in the material	tesla (T)
Permeability μ	$\mu = B/H = \mu_0\mu_r$, with $\mu_0 = 4\pi \times 10^{-7}$ H/m	H/m

Atomic origin of magnetism

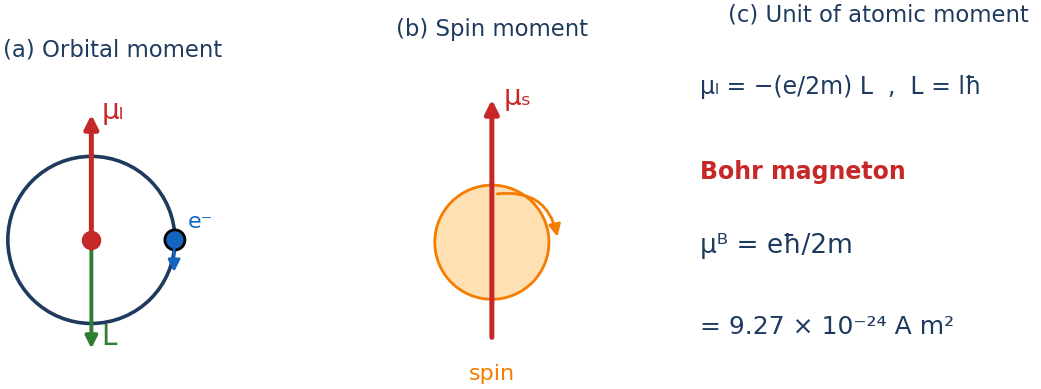


Fig. 6: Orbital motion and electron spin give atoms their magnetic moment

- **Orbital motion** of the electron around the nucleus acts like a tiny current loop and gives an orbital moment $\mu_l = -(e/2m)L$.
- **Electron spin** gives a spin moment of about one Bohr magneton; it is the main source in Fe, Co, Ni.
- **Nuclear spin** moments are about 10^3 times smaller and negligible.
- In filled shells the moments cancel in pairs (Pauli principle), so only atoms with **unpaired electrons** carry a net moment.

6 Classification of Magnetic Materials

Property	Dia	Para	Ferro	Antiferro	Ferri
Permanent atomic moment	None	Yes	Yes	Yes	Yes
Interaction	Induced only	Negligible	Strong, parallel (exchange)	Antiparallel, equal	Antiparallel, unequal
χ	Small, −ve	Small, +ve	Very large, +ve	Small, +ve	Large, +ve
μ_r	< 1	≈ 1	≫ 1	≈ 1	≫ 1
Temperature effect	None	$\chi = C/T$	Becomes para above T_c	Becomes para above T_N	Para above T_c
Example	Cu, Bi	Al, Pt	Fe, Co, Ni	MnO, Cr	Fe_3O_4

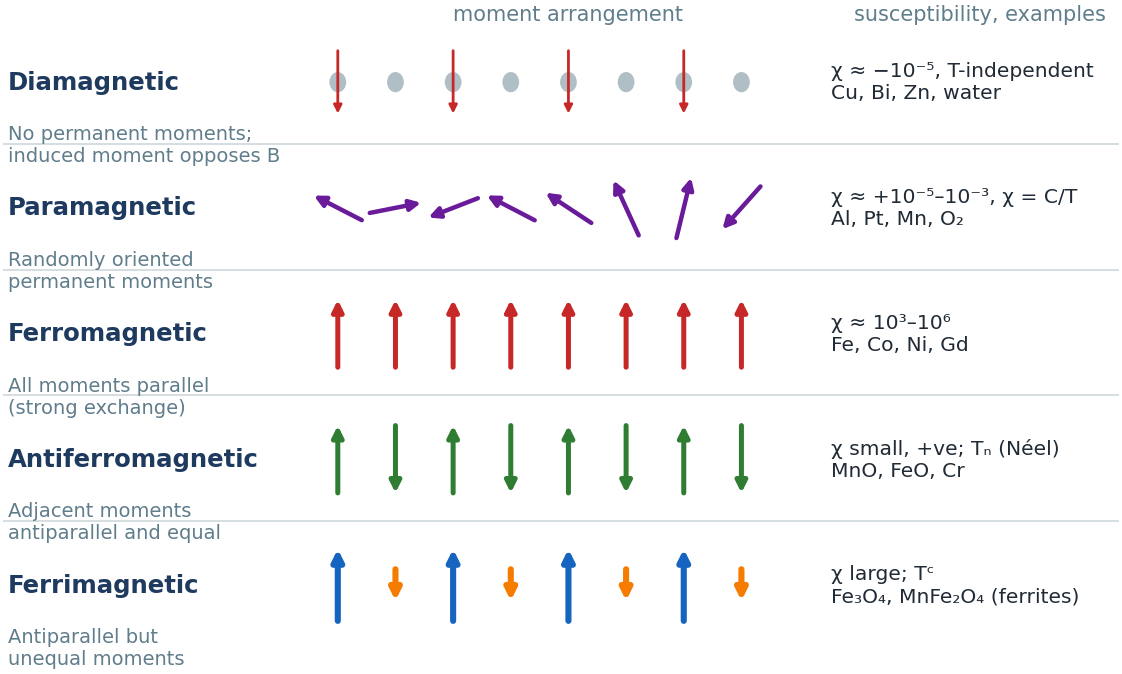


Fig. 7: Moment arrangement in dia-, para-, ferro-, antiferro- and ferrimagnetic materials

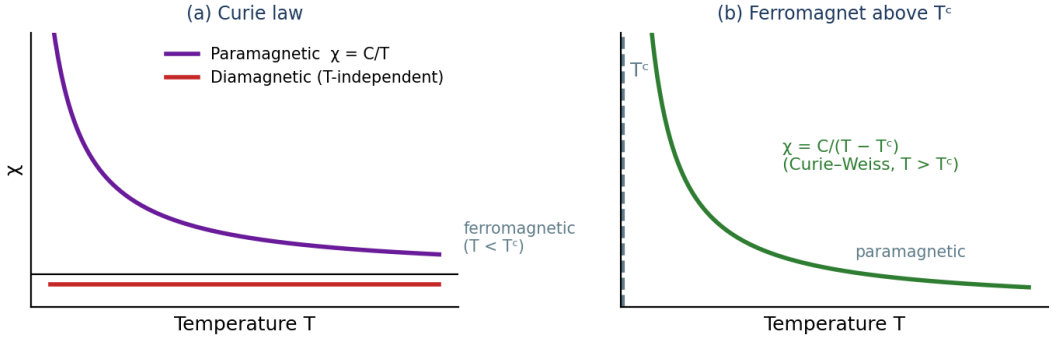


Fig. 8: Temperature dependence: Curie law and Curie–Weiss law

Curie law (paramagnet)	$\chi = \frac{C}{T}$
Curie–Weiss law (ferromagnet, $T > T^c$)	$\chi = \frac{C}{T - T_c}$

Curie temperatures: Fe 1043 K, Co 1388 K, Ni 627 K, Gd 292 K.

7 Domain Theory of Ferromagnetism and Hysteresis

Weiss proposed that a ferromagnet is divided into tiny regions, **domains** (about 10^{-6}–10^{-3} cm), each magnetised to saturation by exchange interaction. Without a field the domains point in different directions, so the net magnetisation is zero.

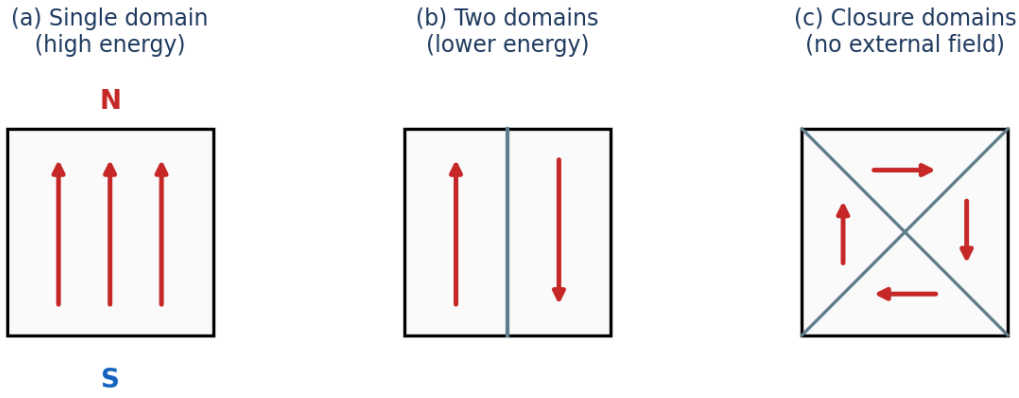
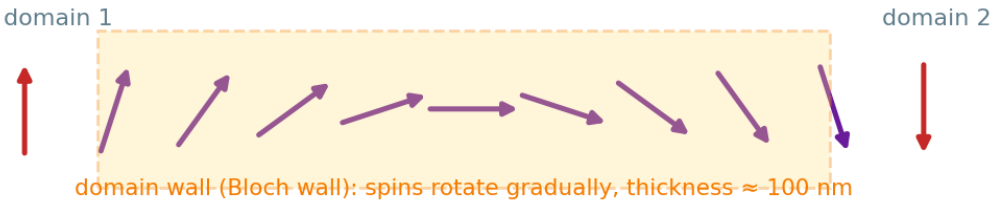


Fig. 9: Domains reduce magnetostatic energy: (a) single domain, (b) two, (c) closure domains with zero external field

DEFINITION • Domain wall (Bloch wall)
The transition region (~100 nm thick) between neighbouring domains in which the spin direction rotates gradually (by 180° or 90°). Gradual rotation lowers exchange energy, while wall thickness is limited by anisotropy energy.



- **Weak field:** domains already aligned with H grow by **wall displacement** (reversible).
- **Stronger field:** irreversible wall motion, then **domain rotation** aligns the rest; the material saturates.
- Domain structure is set by four energies: exchange, magnetostatic, anisotropy and magnetostrictive energy.

Hysteresis

When H is cycled, B lags behind H and traces a closed loop: **hysteresis**.

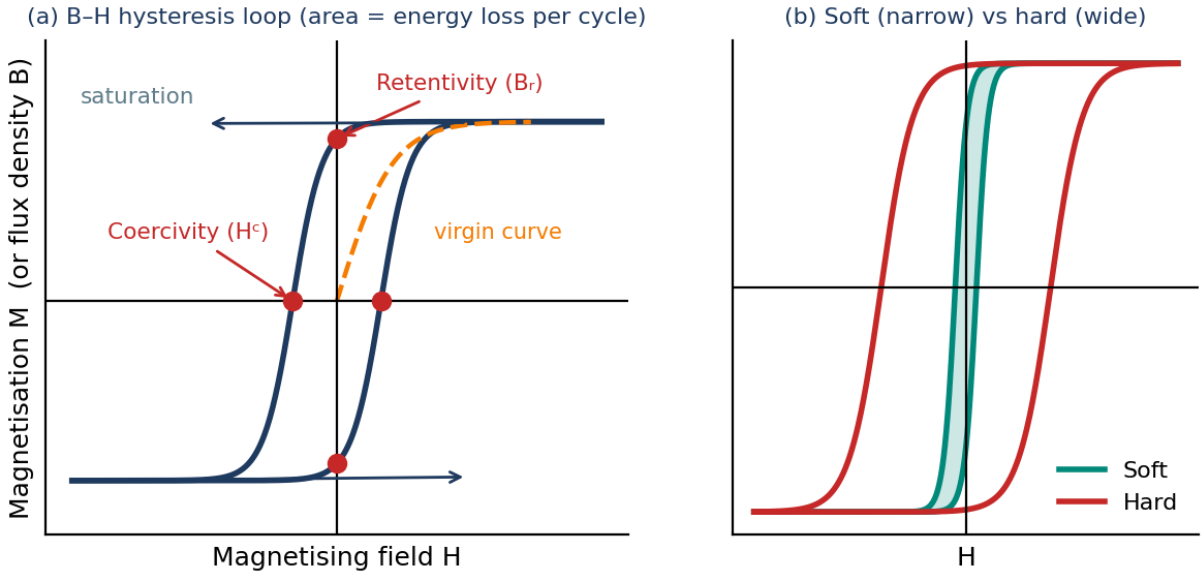


Fig. 10: Hysteresis loop and the comparison of soft and hard magnets

- **Retentivity (B_r):** flux density left when H returns to zero.
- **Coercivity (H_c):** reverse field needed to bring B to zero.
- **Loop area** = energy lost as heat per unit volume per cycle (hysteresis loss).

Property	Soft magnetic materials	Hard magnetic materials
Loop	Narrow, tall, small area	Wide, large area
Coercivity H_c, retentivity	Low H_c	High H_c and high B_r
Permeability	High	Low
Hysteresis loss	Small	Large
Magnetise / demagnetise	Easy	Hard
Materials	Silicon steel, permalloy, soft ferrites	Alnico, NdFeB, SmCo, hard ferrites
Uses	Transformer cores, motors, electromagnets, inductors	Permanent magnets, loudspeakers, magnetic storage

KEY IDEA • Exchange interaction (Weiss field)
Ferromagnetism needs a strong quantum **exchange interaction** between neighbouring spins that favours parallel alignment (equivalent to an internal "Weiss molecular field" of about 10^3 T). Above T_c thermal energy beats exchange and the order disappears.

APPLICATIONS • Applications of magnetic materials

- **Soft:** transformer and motor cores (silicon steel), inductors and high-frequency cores (soft ferrites), magnetic shielding (permalloy).
- **Hard:** permanent magnets in loudspeakers, motors, MRI, generators (NdFeB, SmCo, alnico).
- **Ferrites:** high resistivity → low eddy-current loss at microwave frequencies; memory cores, magnetic tapes and hard disks.

APPLICATIONS • Applications of dielectrics

- Capacitors (high ε_r ceramics like $BaTiO_3$ give large capacitance), insulation of cables, transformer oil.
- Piezoelectric and ferroelectric devices: sensors, ultrasound transducers, memory.
- Microwave resonators and substrates (low-loss ceramics).

8 Worked Problems

EXAMPLE • 1. Electronic polarisability

For an atom of radius R = 1.5 Å: $\alpha_e = 4\pi\varepsilon_0 R^3 = 1.113\times10^{-10} \times (1.5\times10^{-10})^3 = \mathbf{3.76 \times 10^{-40}\ F\cdot m^2}$.

EXAMPLE • 2. Loss tangent

A dielectric has $\varepsilon' = 5.0$ and $\varepsilon'' = 0.01$. $\tan\delta = 0.01/5.0 = \mathbf{2 \times 10^{-3}}$; at 1 MHz and $E_{rms} = 10^4$ V/m the loss is $P = \omega\varepsilon_0\varepsilon''E^2 = 2\pi\times10^6 \times 8.85\times10^{-12} \times 0.01 \times 10^8 \approx \mathbf{55.6\ W/m^3}$.

EXAMPLE • 3. Bohr magneton

$\mu_B = e\hbar/2m = (1.6\times10^{-19} \times 1.055\times10^{-34}) / (2 \times 9.11\times10^{-31}) = \mathbf{9.27 \times 10^{-24}\ A\cdot m^2}$.

EXAMPLE • 4. Curie–Weiss

Iron ($T_c = 1043$ K) has $\chi = 5 \times 10^{-3}$ at 1100 K. $C = \chi(T - T_c) = 5\times10^{-3} \times 57 = 0.285$ K; at 1200 K, $\chi = 0.285/157 = \mathbf{1.8 \times 10^{-3}}$.

Key terms at a glance

Term	Meaning
Bound charge	Charge held to its atom, can only shift slightly
Relaxation time τ	Time for polarisation to reach 1/e of its final value after a field change
Exchange energy	Quantum energy that makes neighbouring spins parallel (ferro) or antiparallel (antiferro)
Magnetostriction	Change of dimensions of a magnetic material when it is magnetised
Eddy current	Circulating current induced in a conductor by changing flux, causing heat loss

9 Formula Sheet and Exam Corner

Topic	Key formulae
Electric vectors	$D = \varepsilon_0 E + P = \varepsilon_0\varepsilon_r E$; $P = \varepsilon_0\chi_e E$; $\varepsilon_r = 1 + \chi_e$
Polarisabilities	$\alpha_e = 4\pi\varepsilon_0 R^3$; $\alpha_i = e^2(1/\beta_1 + 1/\beta_2)$; $\alpha_o = \mu^2/3kT$
Lorentz field	$E_i = E + P/3\varepsilon_0$
Clausius–Mossotti	$(\varepsilon_r - 1)/(\varepsilon_r + 2) = N\alpha/3\varepsilon_0$
Loss	$\varepsilon^* = \varepsilon' - j\varepsilon''$; $\tan\delta = \varepsilon''/\varepsilon'$
Magnetic	$B = \mu_0(H + M)$; $\chi = M/H$; $\mu_r = 1 + \chi$; $\mu_B = e\hbar/2m = 9.27\times10^{-24}$ A·m²
Laws	Curie: $\chi = C/T$; Curie–Weiss: $\chi = C/(T - T_c)$

EXAMPLE • Numerical

A paramagnetic material ($\chi = 2.3 \times 10^{-5}$) is in $H = 10^6$ A/m. $M = \chi H = \mathbf{23\ A/m}$; $B = \mu_0(H + M) = 4\pi\times10^{-7} \times (10^6 + 23) \approx \mathbf{1.2566\ T}$.

EXAM CORNER • Short questions (2–3 marks)

- Define dielectric polarisation, polarisability and susceptibility.
- Write the relation among D, E and P.
- What are the different types of polarisation?
- What is the Lorentz internal field? Write the Clausius–Mossotti equation.
- Define dielectric loss and loss tangent.
- Define Bohr magneton.
- Distinguish between soft and hard magnetic materials.
- What is a domain wall?

EXAM CORNER • Long questions (10 marks)

- Derive expressions for electronic and ionic polarisability.
- Explain orientation polarisation and its dependence on temperature.
- Derive the Clausius–Mossotti equation using the Lorentz internal field.
- Discuss the frequency dependence of polarisation and dielectric loss.
- Classify magnetic materials with the help of neat diagrams and examples.
- Explain the domain theory of ferromagnetism and the hysteresis loop.

EXAM CORNER • Quick MCQs

- 1. Which polarisation depends on temperature? (a) electronic (b) ionic (c) orientation.
- 2. Local field in a cubic crystal: (a) $E + P/3\varepsilon_0$ (b) $E - P/3\varepsilon_0$ (c) E.
- 3. Hard magnets have (a) low H_c (b) high H_c and B_r.
- 4. Antiferromagnets become paramagnetic above (a) Curie (b) Néel temperature.
- **Answers:** 1-c, 2-a, 3-b, 4-b.

QUICK SUMMARY

- Dielectric: $D = \varepsilon_0 E + P$; polarisation = electronic + ionic + orientation + space charge, each cutting off at its own frequency.
- Clausius–Mossotti connects ε_r to α using the Lorentz field $E + P/3\varepsilon_0$.
- Magnetism comes from electron orbit and spin; domains and hysteresis explain ferromagnets, and soft/hard materials suit different devices.

UNIT

IV

ENGINEERING PHYSICS • DIGITAL NOTES

Quantum Mechanics and Free Electron Theory

Matter waves • Uncertainty principle • Wave function • Schrödinger equations • Particle in a box • Classical and quantum free electron theory • Fermi–Dirac • Density of states • Fermi energy

QUICK SUMMARY • What you will learn

- Why particles behave like waves and what that means for what we can know about them
- How the Schrödinger equation gives quantised energies (particle in a box)
- How free electron theories explain conduction in metals, and Fermi–Dirac statistics, density of states and Fermi energy

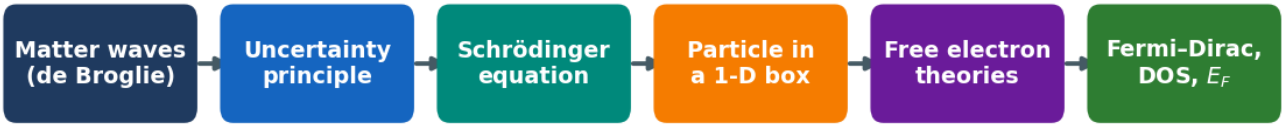


Fig. 1: Roadmap of Unit IV

1 Dual Nature of Matter

DEFINITION • de Broglie hypothesis (1924)

Every moving particle has an associated wave (matter wave) of wavelength λ = h/p, where p = mv is its momentum and h = 6.626 × 10^{-34} J·s.

de Broglie wavelength	$\lambda = \frac{h}{mv} = \frac{h}{p}$
Particle of kinetic energy E	$\lambda = \frac{h}{\sqrt{2mE}}$
Electron accelerated through V volts	$\lambda = \frac{h}{\sqrt{2meV}} = \frac{12.27}{\sqrt{V}}$ AA

EXAMPLE • Numerical

An electron accelerated through 100 V: λ = 12.27/√100 = **1.227 Å**, comparable to atomic spacing, so crystals can diffract electrons. A 0.15 kg cricket ball at 40 m/s has λ = 6.626×10^{-34}/6 = **1.1 × 10^{-34} m**, far too small to observe.

- **Davisson-Germer experiment (1927):** electrons of 54 V scattered from a nickel crystal show a strong peak at φ = 50°.
- Surface diffraction: d sinφ = nλ with d = 2.15 Å gives λ = 1.65 Å, matching the de Broglie value 12.27/√54 = 1.67 Å.
- This confirmed the wave nature of electrons (also G. P. Thomson's electron diffraction).

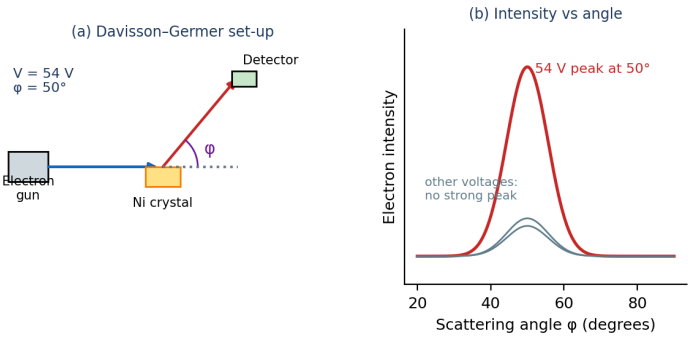


Fig. 2: Davisson–Germer electron diffraction

Properties of matter waves

- Associated with every moving particle, charged or neutral; λ decreases as mass or velocity increases.
- Not electromagnetic waves, they need no medium, and they cannot be observed for ordinary macroscopic bodies.

- Phase velocity $v_p = c^2/v$ (greater than c, no physical significance); group velocity $v_g = v$, equal to the particle velocity.
- The particle is represented by a **wave packet**, a group of waves with slightly different λ.

2 Heisenberg's Uncertainty Principle

DEFINITION • Statement
It is impossible to determine simultaneously and exactly both the position and the momentum of a particle. The product of the uncertainties is at least of the order of ħ/2.

Position–momentum	$\Delta x\,\Delta p \geq \frac{\hbar}{2} \qquad \left(\hbar = \frac{h}{2\pi}\right)$
Energy–time	$\Delta E\,\Delta t \geq \frac{\hbar}{2}$

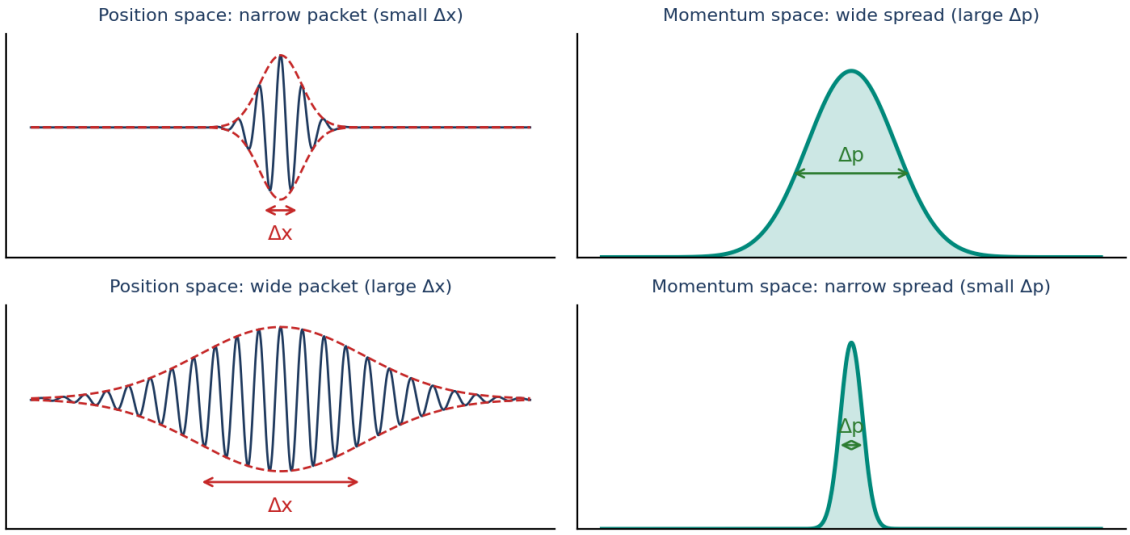


Fig. 3: A localised packet (small Δx) needs many wavelengths (large Δp), and vice versa

EXAMPLE • Can an electron live inside a nucleus?
Nuclear size $\Delta x \approx 10^{-14}$ m → $\Delta p \geq \hbar/2\Delta x = 5.3 \times 10^{-21}$ kg·m/s → $E \approx pc \approx 1.6 \times 10^{-12}$ J ≈ **10 MeV**. Beta electrons emerge with only 1–4 MeV, so electrons cannot exist inside the nucleus.

- **Significance:** it is a property of nature, not a limit of the instrument; it replaces exact orbits by probabilities.
- It explains the natural width of spectral lines (finite lifetime Δt) and why electrons have non-zero minimum (zero-point) energy in an atom or box.

3 Wave Function and Schrödinger Equations

DEFINITION • Wave function Ψ(x, t)
A complex function that describes the quantum state of a particle. It has no direct physical meaning, but $|\Psi|^2 = \Psi\Psi^*$ is the **probability density**: the probability of finding the particle in dx is $|\Psi|^2 dx$ (Born).

Property / condition	Meaning
Single-valued	One probability at every point
Finite and continuous	Probability cannot be infinite or jump
ψ and dψ/dx continuous	Needed for a finite second derivative (except at infinite potentials)
Normalisable	$\int \|\psi\|^2\, dx = 1$ over all space: the particle must be somewhere
Eigenfunction	Allowed solution for a given eigenvalue (energy)

Operators and expectation values (just the idea)

Observable	Operator	Eigenvalue equation
Position x	x (multiply)	—
Momentum p	$-i\hbar\, d/dx$	$p\psi = p\psi$
Energy (Hamiltonian H)	$-(\hbar^2/2m)\, d^2/dx^2 + V$	$H\psi = E\psi$

Probability in the interval (a, b)

$$P = \int_a^b |\psi|^2 dx$$

Average value of x

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi \, dx$$

Time-independent Schrödinger equation

Start with a matter wave of wavelength $\lambda = h/p$ and the classical wave equation for its amplitude: $d^2\psi/dx^2 + (4\pi^2/\lambda^2)\psi = 0$. Since $p^2 = 2m(E - V)$, $\lambda^2 = h^2/2m(E - V)$. Substituting:

1-D

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

3-D

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0 \iff H\psi = E\psi$$

Time-dependent Schrödinger equation

Write $\Psi(x, t) = \psi(x)\, e^{-iEt/\hbar}$. Differentiating with respect to t gives $i\hbar\, \partial\Psi/\partial t = E\Psi$; replacing $E\Psi$ using the time-independent equation:

Time-dependent equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi$$

REMEMBER • When to use which

The **time-independent** form (stationary states) applies when V does not depend on time, e.g. atoms, wells, solids. The **time-dependent** form is the general law of motion for Ψ.

4 Particle in a One-Dimensional Infinite Potential Well

A particle of mass m moves freely along x between rigid walls at x = 0 and x = L: V = 0 for 0 < x < L, V = ∞ elsewhere. Outside the well ψ = 0.



Fig. 4: Solution strategy

Inside the well: $d^2\psi/dx^2 + k^2\psi = 0$ with $k^2 = 2mE/\hbar^2$. General solution $\psi = A \sin kx + B \cos kx$.

- $\psi(0) = 0 \rightarrow B = 0$, so $\psi = A \sin kx$.
- $\psi(L) = 0 \rightarrow \sin kL = 0 \rightarrow kL = n\pi$, n = 1, 2, 3 … (n = 0 gives ψ = 0: no particle).
- Normalisation $\int_0^L A^2 \sin^2(n\pi x/L)\, dx = 1 \rightarrow A^2\, L/2 = 1 \rightarrow A = \sqrt{(2/L)}$.

Eigenfunctions

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\frac{n\pi x}{L}$$

Energy eigenvalues

$$E_n = \frac{n^2h^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

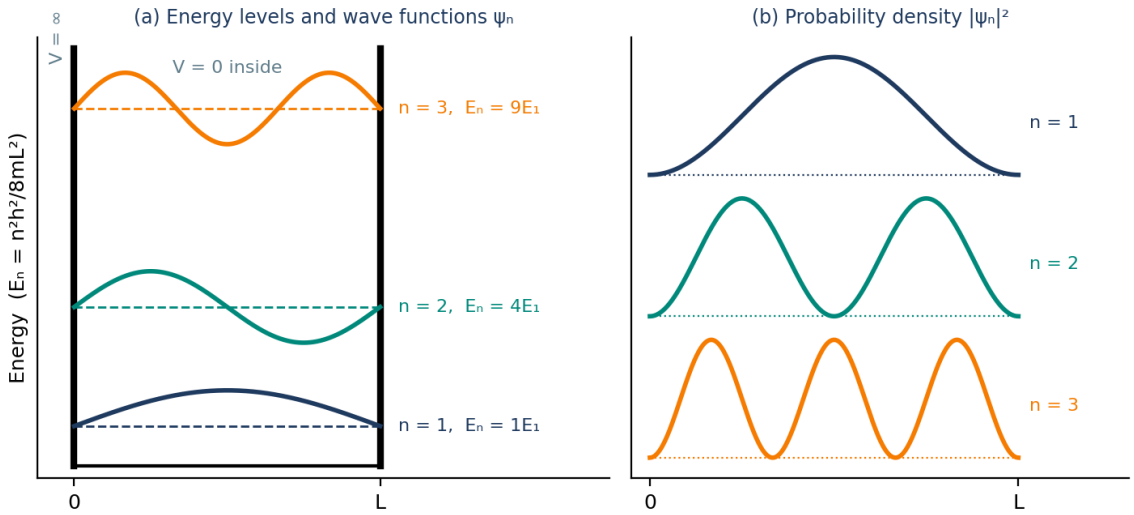


Fig. 5: Energy levels, wave functions and probability densities of a particle in a box

- **Energy is quantised:** only E_1, $4E_1$, $9E_1$ … allowed; spacing grows with n.
- **Zero-point energy** $E_1 = h^2/8mL^2 \neq 0$: the particle can never be at rest (uncertainty principle).
- n − 1 **nodes** inside the well. For n = 2 the probability of being at the centre is zero, while classically it is uniform.
- Smaller L → larger energies (quantum confinement, the basis of quantum dots). For large n the distribution becomes uniform (classical limit).

EXAMPLE • Numerical
Electron in a well of width L = 1 Å: $E_1 = h^2/8mL^2 = (6.626\times10^{-34})^2 / (8 \times 9.11\times10^{-31} \times 10^{-20}) = 6.0\times10^{-18}$ J = **37.6 eV**; $E_2 = 4E_1$ = **150.4 eV**. Photon for 2 → 1: 112.8 eV, λ = 12400/112.8 ≈ **110 Å**.

5 Classical Free Electron Theory (Drude-Lorentz)

Postulates: a metal contains a gas of free valence electrons moving randomly among fixed positive ions; electrons obey the **Maxwell–Boltzmann** classical statistics, and collisions with ions are the only interaction. An electric field adds a slow drift to the random motion.

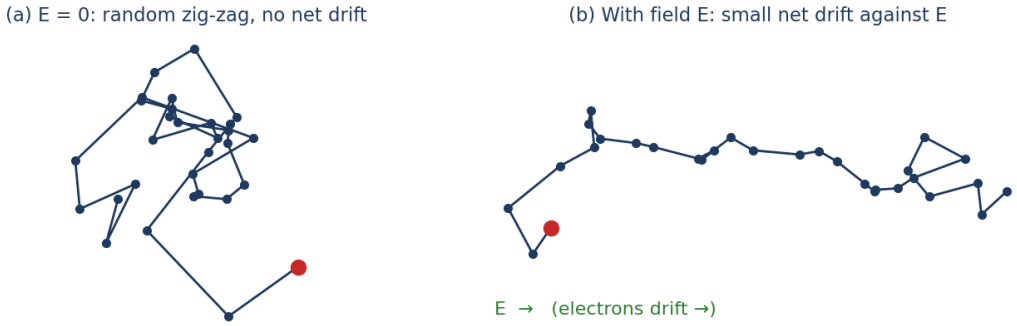


Fig. 6: Random motion of an electron with and without an applied field

Under field E the electron accelerates, a = −eE/m, until a collision. With mean free time (relaxation time) τ, drift velocity $v_d = a\tau$. Current density $J = n e v_d$:

Drift velocity and mobility	$v_d = \frac{eE\tau}{m}, \qquad \mu = \frac{v_d}{E} = \frac{e\tau}{m}$
Electrical conductivity (Ohm's law)	$\sigma = \frac{J}{E} = \frac{ne^2\tau}{m} = ne\mu$
Wiedemann–Franz law	$\frac{K}{\sigma T} = \frac{3}{2}\left(\frac{k}{e}\right)^2 = 1.12 \times 10^{-8}\ \mathrm{W\Omega K^{-2}}$

Merits	Demerits
Verifies Ohm's law	Fails to explain the electronic specific heat: predicts 3k/2 per electron, observed value is ~100 times smaller
Explains electrical and thermal conductivity	Predicts $\rho \propto \sqrt{T}$, but experimentally $\rho \propto T$
Gives the Wiedemann–Franz law (K/σT = constant)	Lorenz number 1.12×10^{-8} disagrees with the measured 2.44×10^{-8}
Explains metallic lustre and opacity	Cannot explain semiconductors, insulators, ferromagnetism, paramagnetism, photoelectric or Compton effect

6 Quantum Free Electron Theory (Sommerfeld)

Keeps the free electron gas picture but treats electrons **quantum mechanically**:

- Electrons move in a region of **constant potential** (inside the metal), a 3-D box, so their energies are quantised.
- Electrons obey the **Pauli exclusion principle**: at most two (spin up and down) per energy level, filled from the bottom up to the Fermi level.
- They follow **Fermi–Dirac statistics** instead of Maxwell–Boltzmann.
- Electrons are waves with λ = h/p; scattering happens only from **imperfections** and lattice vibrations, not from perfect ions.

Under field E the whole Fermi distribution shifts by $\delta k = -eE\tau/\hbar$, but only electrons near E_F can be scattered into empty states and contribute to the current. The result looks like the classical one with effective mass m*:

Conductivity (quantum)
$$\sigma = \frac{ne^2\tau}{m^*} = \frac{ne^2\lambda_F}{m^* v_F}$$

Mobility
$$\mu = \frac{e\tau}{m^*}$$

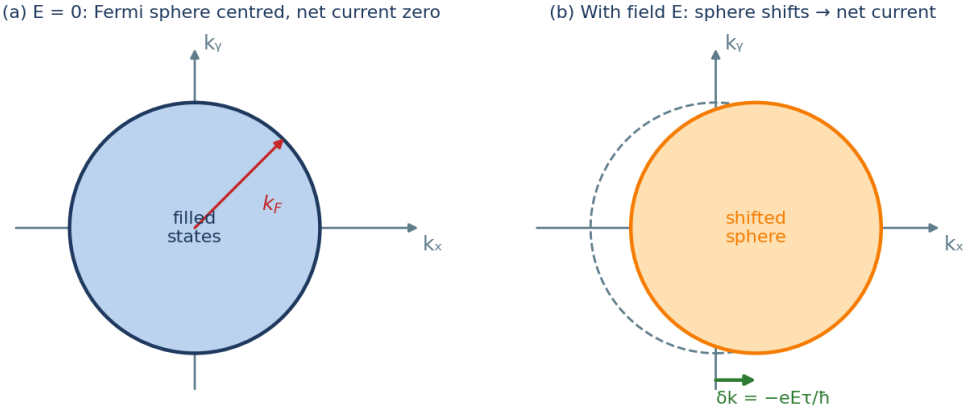


Fig. 7: Fermi sphere in k-space: the field shifts the whole sphere by δk

KEY IDEA • Why conductivity differs between metals
τ (or mean free path λ_F) depends on impurities and temperature; m* depends on how the band curves (Unit V). Resistivity rises with T because lattice vibrations reduce τ, giving $\rho \propto T$ as observed.

Merits of QFET	Demerits of QFET
Explains the small electronic specific heat and the T-independent paramagnetism	Assumes constant potential, ignoring the periodic lattice
Explains $\rho \propto T$ and gives the correct Lorenz number $(\pi^2/3)(k/e)^2 = 2.44\times10^{-8}$	Cannot explain why some solids are conductors, semiconductors or insulators (needs band theory)
Explains thermionic and photoelectric emission	Cannot explain positive Hall coefficients in some metals

7 Fermi–Dirac Distribution, Density of States and Fermi Energy

Fermi–Dirac distribution
$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

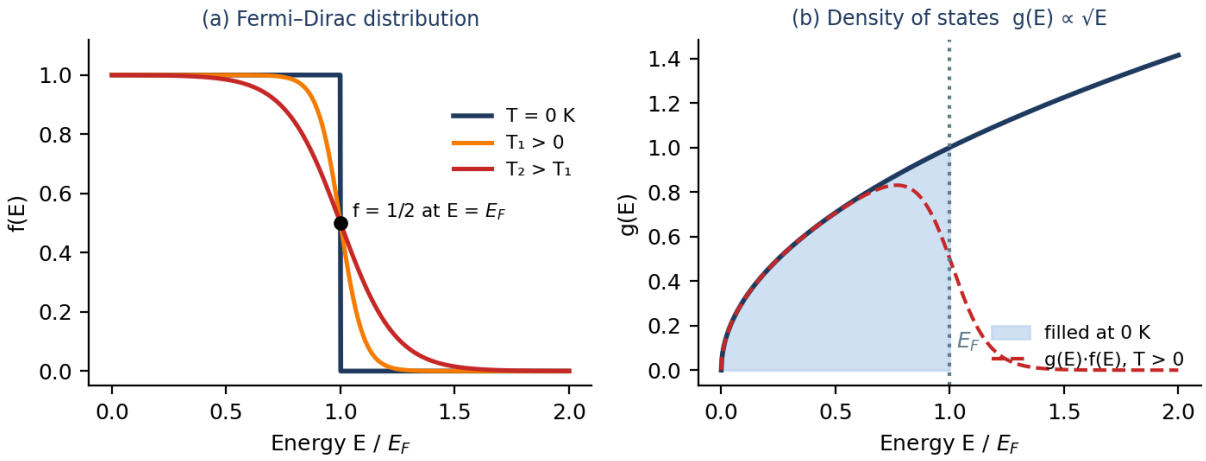


Fig. 8: Fermi–Dirac function and density of states in a metal

- f(E) is the **probability that a level of energy E is occupied** by an electron.
- **T = 0 K:** f = 1 for E < E_F and f = 0 for E > E_F (sharp step).
- **T > 0:** only electrons within ≈ kT of E_F are thermally excited; at E = E_F, f = 1/2 at any temperature.

DEFINITION • Fermi energy (E_F)
The energy of the highest occupied level at 0 K. At T > 0 it is the energy at which the occupation probability is 1/2. Typical values are 2–11 eV for metals (Cu 7.0 eV, Al 11.7 eV, Na 3.2 eV).

Density of states g(E)

g(E) dE is the number of electron states per unit volume between E and E + dE. In a cubic box of side L, $E = (h^2/8mL^2)(n_x^2 + n_y^2 + n_z^2) = n^2h^2/8mL^2$. In n-space the states with energy up to E fill the positive octant of a sphere of radius $n = (8mL^2E/h^2)^{1/2}$; with 2 spin states per level:

States up to E	$N(E) = 2 \times \frac{1}{8} \cdot \frac{4}{3}\pi n^3 = \frac{\pi}{3}\left(\frac{8mL^2E}{h^2}\right)^{3/2}$
Density of states (per unit volume)	$g(E)\,dE = \frac{4\pi}{h^3}(2m)^{3/2}E^{1/2}\,dE$

Fermi energy at 0 K

Electron concentration	$n = \int_0^{E_F} g(E)\,dE = \frac{8\pi}{3h^3}(2mE_F)^{3/2}$
Fermi energy	$E_F = \frac{h^2}{8m}\left(\frac{3n}{\pi}\right)^{2/3} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$
Mean energy at 0 K	$\bar{E} = \frac{3}{5}E_F$

EXAMPLE • Numerical
Copper: $n = 8.47 \times 10^{28}$ m^{-3}. $E_F = (\hbar^2/2m)(3\pi^2 n)^{2/3} = 6.10\times10^{-39} \times 1.84\times10^{20} = 1.13 \times 10^{-18}$ J = **7.03 eV**. Fermi temperature $T_F = E_F/k \approx 8.2 \times 10^4$ K, so at room temperature kT (0.026 eV) ≪ E_F and the electron gas is highly degenerate.

Classical versus quantum free electron theory

Aspect	Classical (Drude–Lorentz)	Quantum (Sommerfeld)
Statistics	Maxwell–Boltzmann	Fermi–Dirac (Pauli principle)
Energy levels	Continuous	Quantised, filled to E_F
Electrons contributing to heat and current	All free electrons	Only those within ~kT of E_F
Electronic specific heat	3k/2 per electron (too large)	$\propto T$, ~1% of classical (agrees with experiment)
Resistivity vs T	$\propto \sqrt{T}$ (wrong)	$\propto T$ (correct)
Electron nature	Particle	Wave, scattered only by imperfections

APPLICATIONS • Where this physics is used

- **Electron microscope:** fast electrons have $\lambda \approx 0.01$–1 Å, far below visible light, giving atomic resolution.
- **Quantum dots and wells:** $E_n \propto 1/L^2$, so size sets the emitted colour (displays, lasers, bio-imaging).
- **Tunnelling devices:** STM, tunnel diodes and flash memory use the leakage of ψ into barriers.
- **Metals and electronics:** E_F and g(E) determine heat capacity, thermionic emission and contact potentials.

REMEMBER • Quantum tunnelling (qualitative)

If a barrier is finite, ψ does not drop to zero at its wall; it decays exponentially inside and can emerge on the other side. A particle can therefore cross a barrier higher than its energy. This is impossible classically, and is used in tunnel diodes, STM and alpha decay.

8 Worked Problems

EXAMPLE • 1. de Broglie wavelength

Find the wavelength of a neutron ($m = 1.675\times10^{-27}$ kg) with kinetic energy 0.025 eV (thermal neutron). $E = 4.0\times10^{-21}$ J; $\lambda = h/\sqrt{(2mE)} = 6.626\times10^{-34}/\sqrt{(2 \times 1.675\times10^{-27} \times 4\times10^{-21})}$ = **1.81 Å**, suitable for diffraction by crystals.

EXAMPLE • 2. Uncertainty principle

The position of an electron is known to within 0.1 Å. $\Delta p \geq \hbar/2\Delta x = 1.055\times10^{-34}/(2 \times 10^{-11}) = 5.3\times10^{-24}$ kg·m/s, so $\Delta v \geq 5.8 \times 10^{6}$ m/s.

EXAMPLE • 3. Particle in a box

Electron in a 2 Å well: $E_1 = 37.6/4$ = **9.4 eV**; $E_3 = 9E_1$ = **84.6 eV**. The probability of finding the electron between x = 0 and L/4 in the ground state is $P = (2/L)\int \sin^2(\pi x/L)dx$ = **0.091**.

EXAMPLE • 4. Conductivity

Silver: $n = 5.86\times10^{28}$ m^{-3}, $\tau = 3.8\times10^{-14}$ s. $\sigma = ne^2\tau/m = 5.86\times10^{28} \times (1.6\times10^{-19})^2 \times 3.8\times10^{-14}/9.11\times10^{-31}$ = **6.3×10^{7} Ω^{-1}m^{-1}**; mobility $\mu = e\tau/m$ = **6.7×10^{-3} m²/Vs**.

EXAMPLE • 5. Fermi function

Probability of occupation at $E = E_F + 0.1$ eV at T = 300 K: $f = 1/(1 + e^{0.1/0.0259}) = 1/(1 + 47.5)$ = **0.0206** (about 2%). At $E = E_F - 0.1$ eV, f = **0.979**.

Key terms at a glance

Term	Meaning
Eigenvalue / eigenfunction	Allowed energy and its wave function, solution of $H\psi = E\psi$
Degenerate states	Different states having the same energy
Relaxation time τ	Average time between two successive collisions of an electron
Drift velocity	Average velocity gained along the field direction
Fermi level / Fermi surface	Highest filled level at 0 K / its boundary in k-space
Effective mass m*	Mass an electron appears to have in a crystal, reflecting the periodic potential

Model answers to common 2-mark questions

KEY IDEA • Why can't we see the wave nature of a cricket ball?
Its momentum is large, so $\lambda = h/p \sim 10^{-34}$ m, far smaller than anything that could act as a diffracting object.

KEY IDEA • Why is the ground-state energy of a particle in a box not zero?
If E = 0 then ψ = 0 everywhere (no particle). Confining the particle to length L makes $\Delta p \geq \hbar/2L$, so it must have kinetic energy of at least about $h^2/8mL^2$.

KEY IDEA • Why does only a small fraction of electrons conduct in a metal?
Pauli exclusion fills all lower states, so only electrons within ~kT of E_F have empty states to move into.

KEY IDEA • What happens to E_F when temperature rises?
For metals it changes very little, because $kT \ll E_F$; only the edge of the distribution is blurred.

EXAM CORNER • Try yourself (answers in brackets)

- Find the de Broglie wavelength of an electron accelerated through 400 V. [0.61 Å]
- A proton and an electron have the same kinetic energy. Which has the shorter wavelength, and by what ratio? [proton; $\sqrt{1836} \approx 43$ times shorter]
- Find E_1 and E_2 for an electron in a 5 Å box. [1.5 eV, 6.0 eV]
- Calculate E_F for aluminium ($n = 18.1 \times 10^{28}$ m^{-3}). [≈ 11.7 eV]
- The lifetime of an excited state is 10^{-8} s. Estimate the minimum energy uncertainty. [$\approx 3.3 \times 10^{-8}$ eV]

9 Formula Sheet and Exam Corner

Topic	Key formulae
de Broglie	$\lambda = h/p = h/\sqrt{(2mE)}$; electrons: $\lambda = 12.27/\sqrt{V}$ Å
Uncertainty	$\Delta x \Delta p \geq \hbar/2$; $\Delta E \Delta t \geq \hbar/2$
Schrödinger	$d^2\psi/dx^2 + (2m/\hbar^2)(E - V)\psi = 0$; $i\hbar\partial\Psi/\partial t = H\Psi$
Particle in box	$\psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$; $E_n = n^2h^2/8mL^2$
Classical FET	$\sigma = ne^2\tau/m$; $\mu = e\tau/m$; $K/\sigma T = (3/2)(k/e)^2$
Quantum FET	$\sigma = ne^2\tau/m^*$
Fermi	$f(E) = 1/[1 + e^{(E-E_F)/kT}]$; $g(E) = (4\pi/h^3)(2m)^{3/2}E^{1/2}$; $E_F = (h^2/8m)(3n/\pi)^{2/3}$

EXAM CORNER • Short questions (2–3 marks)

- State de Broglie's hypothesis. Derive λ for an electron accelerated through V volts.
- State Heisenberg's uncertainty principle.
- What is the physical significance of the wave function?
- Write the conditions a wave function must satisfy.
- Write the Schrödinger time-independent equation.
- Define Fermi energy and Fermi level.
- State the Wiedemann–Franz law.

EXAM CORNER • Long questions (10 marks)

- Derive the Schrödinger time-independent and time-dependent wave equations.
- Solve Schrödinger's equation for a particle in a 1-D infinite potential well and obtain energy eigenvalues and eigenfunctions.
- Explain the classical free electron theory. Derive an expression for electrical conductivity and discuss its merits and demerits.
- Discuss quantum free electron theory and derive $\sigma = ne^2\tau/m^*$.
- Explain the Fermi–Dirac distribution function and its variation with temperature.
- Derive the density of states and hence the Fermi energy at 0 K.

EXAM CORNER • Quick MCQs

- 1. Zero-point energy of a particle in a box is (a) 0 (b) $h^2/8mL^2$ (c) $h^2/2mL^2$.
- 2. At $E = E_F$, f(E) at $T > 0$ is (a) 0 (b) 1/2 (c) 1.
- 3. g(E) is proportional to (a) E (b) $\sqrt{E}$ (c) 1/E.
- 4. Classical theory predicts resistivity $\propto$ (a) T (b) $\sqrt{T}$ (c) 1/T.
- **Answers:** 1-b, 2-b, 3-b, 4-b.

QUICK SUMMARY

- Matter has wave nature ($\lambda = h/p$); $\Delta x \Delta p \geq \hbar/2$.
- Schrödinger equation in a box gives $E_n = n^2h^2/8mL^2$ and $\psi_n = \sqrt{(2/L)} \sin(n\pi x/L)$.
- Classical theory (Maxwell–Boltzmann) gets Ohm's law but fails for specific heat; quantum theory (Fermi–Dirac, Pauli) fixes this: $g(E) \propto \sqrt{E}$, $E_F = (h^2/8m)(3n/\pi)^{2/3}$.

UNIT

V

ENGINEERING PHYSICS • DIGITAL NOTES

Semiconductors

Energy bands • Classification of solids • Intrinsic semiconductors • Extrinsic (n and p type) • Fermi level • Drift and diffusion • Einstein relation • Hall effect and applications

QUICK SUMMARY • What you will learn

- How energy bands form when atoms come together and why that sorts solids into conductors, semiconductors and insulators
- How to find carrier density, Fermi level and conductivity in intrinsic and doped semiconductors
- How currents flow (drift, diffusion) and how the Hall effect tells us the type and density of carriers

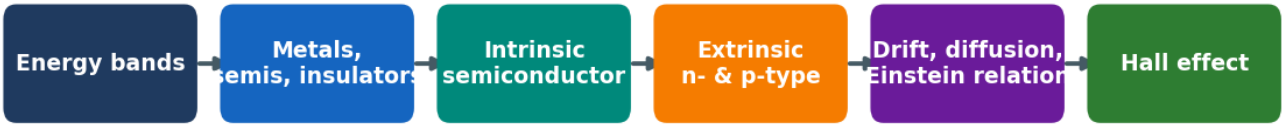


Fig. 1: Roadmap of Unit V

1 Formation of Energy Bands

An isolated atom has sharp, discrete energy levels. When N identical atoms are brought together, the electron clouds overlap, and the **Pauli exclusion principle** forbids two electrons from having the same state. Each atomic level therefore **splits into N closely spaced levels**, which merge into a **continuous band** ($N \approx 10^{23}$).

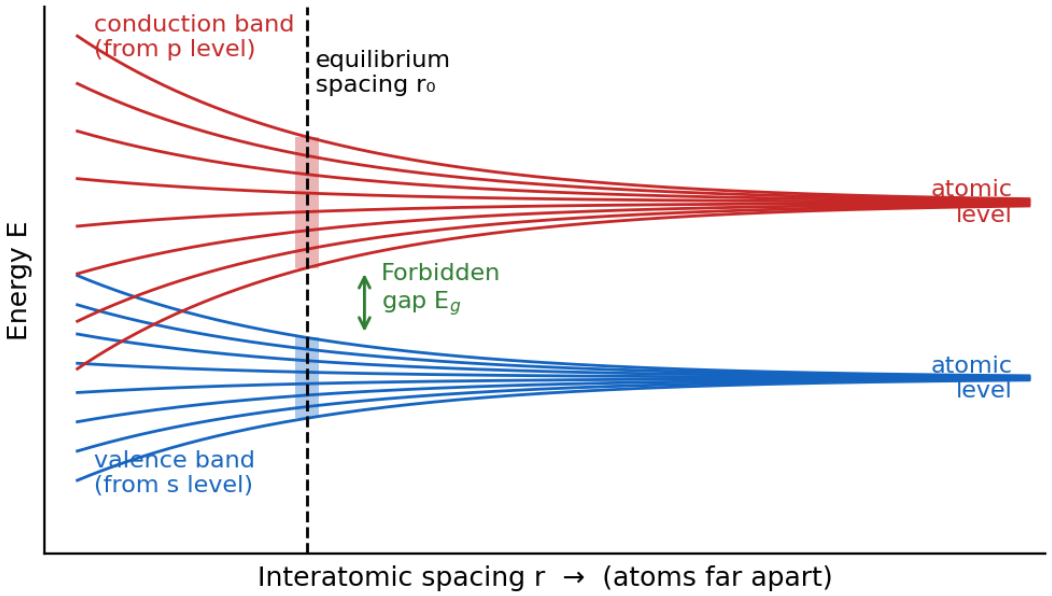


Fig. 2: Splitting of atomic levels into bands as the interatomic spacing decreases

- Outer (valence) electron levels spread most (strong overlap); inner levels stay narrow.
- **Valence band (VB):** highest band filled with valence electrons. **Conduction band (CB):** next higher band, empty or partly filled.
- **Forbidden gap (E_g):** energy range between the top of VB and bottom of CB with no allowed levels.
- The band picture is the key that sorts all solids by how easily electrons can reach empty states.

DEFINITION • Hole and effective mass

A **hole** is the vacancy left in the almost full valence band. It behaves as a particle of charge +e. The **effective mass m*** packages the effect of the crystal's periodic potential on a carrier: $m^* = \hbar^2/(d^2E/dk^2)$. Near band edges the carrier then behaves like a free particle of mass m*.

REMEMBER • Kronig–Penney idea (qualitative)
An electron in the periodic potential of a lattice has allowed energy bands separated by forbidden gaps because waves with k = nπ/a are Bragg-reflected and form standing waves with two different energies, leaving a gap between them.

2 Classification of Crystalline Solids

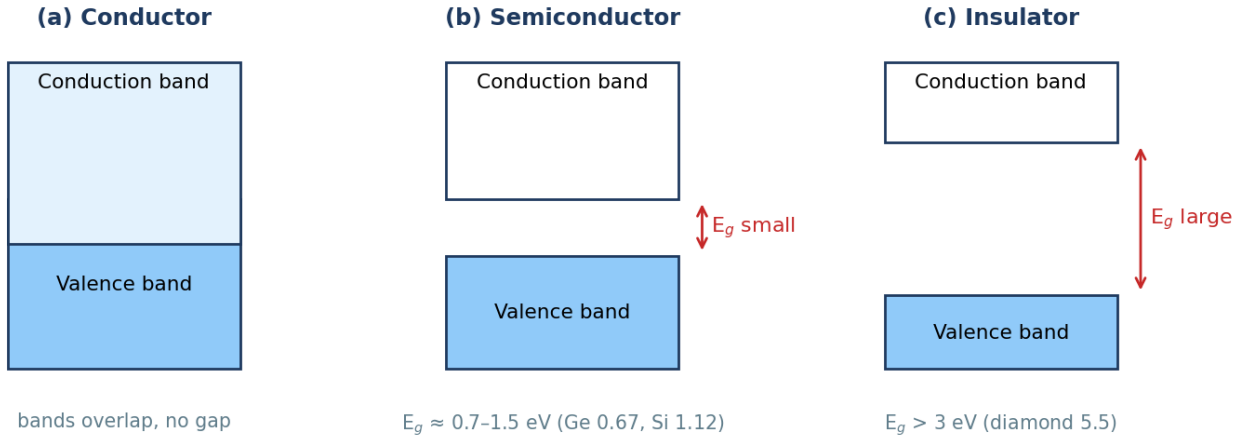


Fig. 3: Band diagrams of a conductor, a semiconductor and an insulator

Property	Conductor (metal)	Semiconductor	Insulator
Band gap E_g	None (bands overlap or CB part filled)	≈ 0.1–3 eV	> 3 eV
Resistivity (Ω·m)	10^{-8}–10^{-6}	10^{-4}–10^{6}	10^{8}–10^{16}
Free carriers at 300 K	Very many (~10^{28} m^{-3})	Few (~10^{16}–10^{19} m^{-3})	Almost none
Effect of rising T	ρ increases (more scattering)	ρ decreases (more carriers)	ρ decreases, but stays high
Examples	Cu, Al, Ag	Si (1.12 eV), Ge (0.67 eV), GaAs (1.43 eV)	Diamond (5.5 eV), glass, mica

KEY IDEA • Temperature coefficient
At 0 K a pure semiconductor is an insulator: VB full, CB empty. Thermal energy kT (0.026 eV at 300 K) lifts a few electrons across E_g leaving **holes** behind, which behave as positive carriers.

3 Intrinsic Semiconductors

DEFINITION • Intrinsic semiconductor
A pure semiconductor (no impurities) in which electrons in the CB and holes in the VB are produced only by thermal generation of electron\u2013hole pairs, so n = p = n_i.

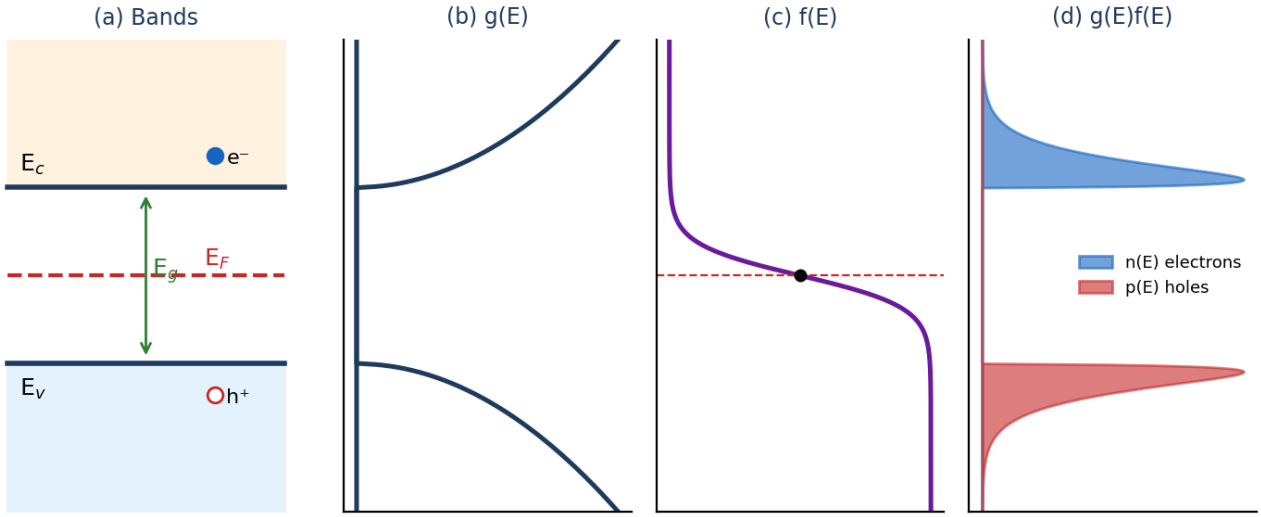


Fig. 4: Bands, density of states g(E), Fermi function f(E) and the resulting carrier distributions

Density of charge carriers

Electron density in the CB: $n = \int g_c(E)\, f(E)\, dE$ from E_c upward, with $g_c(E) = (4\pi/h^3)(2m_e^*)^{3/2}(E - E_c)^{1/2}$. Since $E - E_F \gg kT$ (non-degenerate), $f(E) \approx e^{-(E-E_F)/kT}$. Integrating:

Electrons in CB	$n = N_c\, e^{-(E_c - E_F)/kT}, \qquad N_c = 2\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/2}$
Holes in VB (probability $1 - f$)	$p = N_v\, e^{-(E_F - E_v)/kT}, \qquad N_v = 2\left(\frac{2\pi m_h^* kT}{h^2}\right)^{3/2}$
Intrinsic: $n = p = n_i$	$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}(m_e^* m_h^*)^{3/4}\, e^{-E_g/2kT}$
Law of mass action	$n\,p = n_i^2 = N_c N_v\, e^{-E_g/kT}$

Fermi level

KEY IDEA • Derivation in three lines

- $n = p \rightarrow N_c\, e^{-(E_c - E_F)/kT} = N_v\, e^{-(E_F - E_v)/kT}$.
- Take logarithms: $2E_F = E_c + E_v + kT \ln(N_v/N_c)$.
- $N_v/N_c = (m_h^*/m_e^*)^{3/2} \rightarrow E_F = (E_c + E_v)/2 + (3kT/4)\ln(m_h^*/m_e^*)$.

Setting $n = p$ and solving for E_F:

Intrinsic Fermi level	$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4}\ln\frac{m_h^*}{m_e^*}$

If $m_h^* \approx m_e^*$ (or at T = 0 K), E_F lies in the **middle of the gap**, at $E_g/2$ above the VB. Because m_h^* is a bit larger than m_e^*, E_F drifts slightly upward with T.

Electrical conductivity

Conductivity	$\sigma = ne\mu_e + pe\mu_h = n_i e\,(\mu_e + \mu_h) \;\propto\; T^{3/2} e^{-E_g/2kT}$
Straight-line test	$\ln\sigma = \text{const} - \frac{E_g}{2k}\cdot\frac{1}{T} \;\Rightarrow\; \text{slope} = -\frac{E_g}{2k}$

EXAMPLE • Numerical

Germanium at 300 K: $n_i = 2.4 \times 10^{19}\ m^{-3}$, $\mu_e = 0.39$ and $\mu_h = 0.19\ m^2/Vs$. $\sigma = n_i e(\mu_e + \mu_h) = 2.4\times10^{19} \times 1.6\times10^{-19} \times 0.58 =$ **2.2 (Ωm)$^{-1}$**.

APPLICATIONS • Why semiconductors matter

- Diodes, transistors and ICs: controlled conductivity by doping.
- Solar cells and photodetectors: photons of energy ≥ E_g create electron–hole pairs.
- LEDs and laser diodes: direct-gap materials (GaAs, GaN) emit light on recombination; Si is indirect and a poor emitter.
- Thermistors, Hall sensors, gas sensors.

4 Extrinsic Semiconductors

Doping adds a controlled trace (1 part in 10^6–10^9) of impurity atoms. Result: an **extrinsic semiconductor** whose conductivity is set mostly by the dopant.

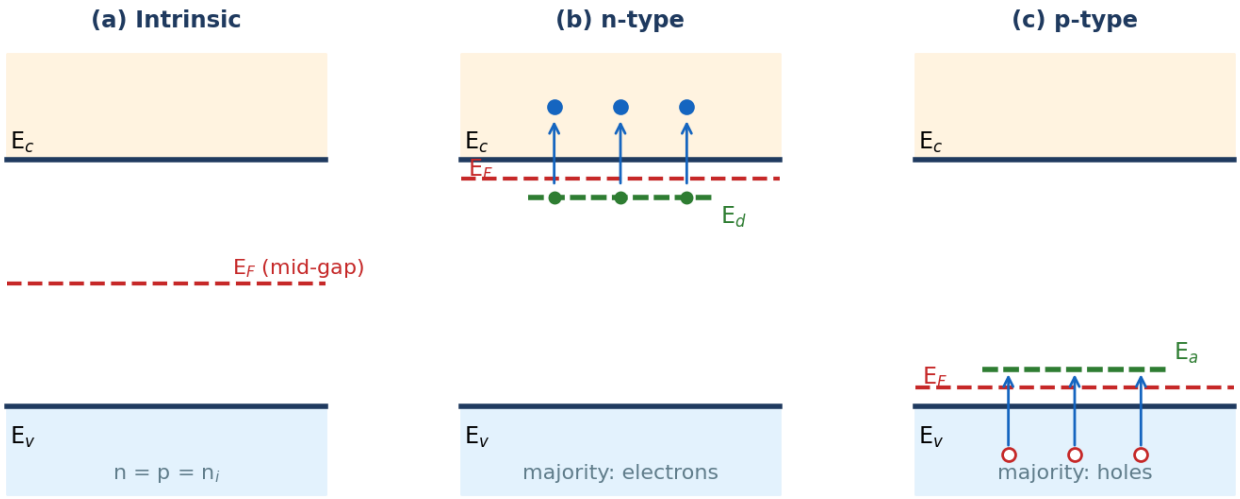


Fig. 5: Energy bands of intrinsic, n-type and p-type semiconductors

	n-type	p-type
Dopant	Pentavalent donor (P, As, Sb)	Trivalent acceptor (B, Al, Ga, In)
Extra level	Donor level E_d just **below** E_c (~0.01–0.05 eV)	Acceptor level E_a just **above** E_v
What happens	Fifth electron is easily freed to the CB; donor becomes a fixed positive ion	Accepts an electron from VB; leaves a mobile hole and a fixed negative ion
Majority / minority	Electrons / holes	Holes / electrons
At 0 K, E_F	Midway between E_d and E_c	Midway between E_v and E_a

Feature	Intrinsic	Extrinsic
Purity	Pure crystal	Doped (10^{-6}–10^{-9})
Carriers	$n = p = n_i$	$n \neq p$; majority + minority
Conductivity	Low, strongly T-dependent	Higher and nearly T-independent in the working range
Fermi level	Mid-gap	Near E_c (n) or E_v (p)
Use	Reference and theory	All devices

Carrier density in extrinsic semiconductors

n-type (low T, $\Delta E = E^c - E^d$)

$$n = (2N_d)^{1/2}\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/4} e^{-\Delta E/2kT}$$

p-type (low T, $\Delta E = E_a - E_v$)

$$p = (2N_a)^{1/2}\left(\frac{2\pi m_h^* kT}{h^2}\right)^{3/4} e^{-\Delta E/2kT}$$

Mass action (any T)

$$np = n_i^2 \Rightarrow p_n = \frac{n_i^2}{N_d}, \quad n_p = \frac{n_i^2}{N_a}$$

Conductivity

$$\sigma_n \approx N_d e\mu_e, \qquad \sigma_p \approx N_a e\mu_h$$

Dependence of Fermi energy on concentration and temperature

n-type, near 0 K

$$E_F = \frac{E_c + E_d}{2} + \frac{kT}{2}\ln\frac{N_d}{2(2\pi m_e^* kT/h^2)^{3/2}}$$

p-type, near 0 K

$$E_F = \frac{E_v + E_a}{2} - \frac{kT}{2}\ln\frac{N_a}{2(2\pi m_h^* kT/h^2)^{3/2}}$$

After full ionisation

$$E_F - E_i = kT\ln\frac{N_d}{n_i}\ \text{(n-type)}, \quad E_i - E_F = kT\ln\frac{N_a}{n_i}\ \text{(p-type)}$$

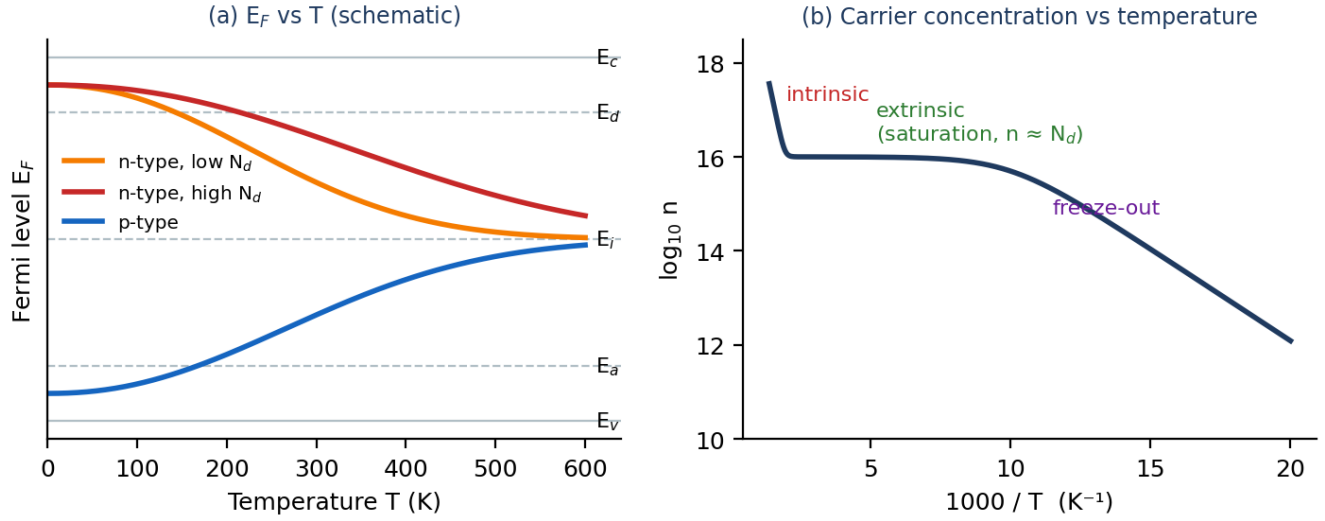


Fig. 6: Fermi level versus temperature and carrier concentration versus 1/T (schematic)

- **Concentration:** more N_d pushes E_F closer to E_c (n-type); more N_a pushes it closer to E_v (p-type).
- **Temperature:** as T rises, E_F moves from near the band edge toward the **intrinsic level E_i** (mid-gap). Heavier doping needs a higher temperature before the semiconductor turns intrinsic.
- Three regions in n(T): **freeze-out** (low T, donors not all ionised) → **saturation** ($n \approx N_d$, device range) → **intrinsic** (high T, $n \approx n_i$).

EXAMPLE • Numerical
Silicon doped with $N_d = 10^{22}$ m^{-3} at 300 K ($n_i = 1.5 \times 10^{16}$ m^{-3}): $n \approx 10^{22}$ m^{-3}; $p = n_i^2/N_d$ = **2.25×10^{10} m^{-3}**; $E_F - E_i = kT \ln(N_d/n_i) = 0.0259 \times 13.4$ = **0.347 eV** above mid-gap.

5 Drift and Diffusion Currents; Einstein's Relation

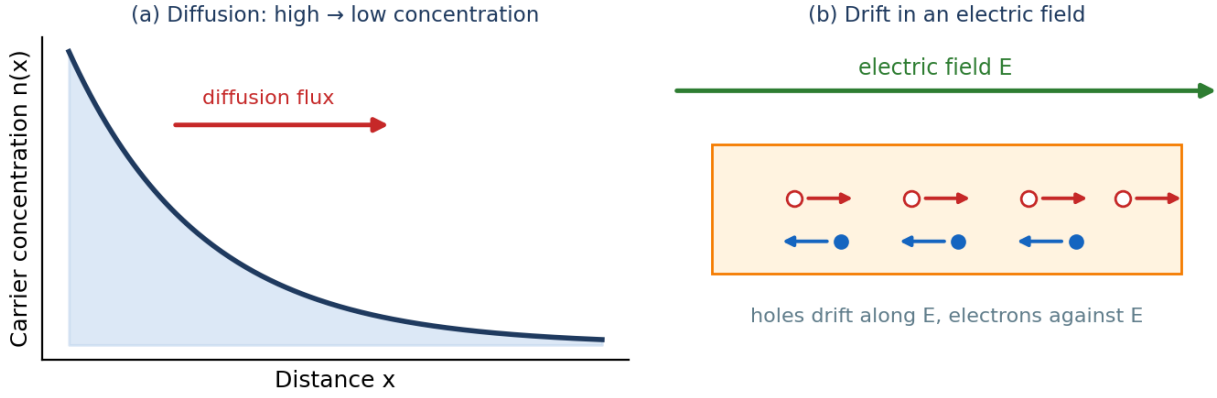


Fig. 7: Diffusion (concentration gradient) and drift (electric field)

	Drift current	Diffusion current
Cause	Electric field E	Concentration gradient dn/dx or dp/dx
Mobility / diffusion	Mobility $\mu = v_d/E$ (m²/Vs)	Diffusion coefficient D (m²/s)
Electrons	$J_n = n\,e\,\mu_n\,E$	$J_n = e\,D_n\,(dn/dx)$
Holes	$J_p = p\,e\,\mu_p\,E$	$J_p = -e\,D_p\,(dp/dx)$

Total current density $$J = J_n + J_p = e(n\mu_n + p\mu_p)E + e\left(D_n\frac{dn}{dx} - D_p\frac{dp}{dx}\right)$$

Einstein relation $$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$$

Diffusion length $$L = \sqrt{D\tau}$$

KEY IDEA • Proof of the Einstein relation (idea)

- In equilibrium the net current is zero: $n\mu E + D\, dn/dx = 0$.
- Boltzmann distribution in potential V(x): $n = n_0\, e^{eV/kT} \rightarrow dn/dx = -(e/kT)\, nE$.
- Substitute: $n\mu E - D(e/kT)nE = 0 \rightarrow$ **$D/\mu = kT/e$** (= 0.0259 V at 300 K).

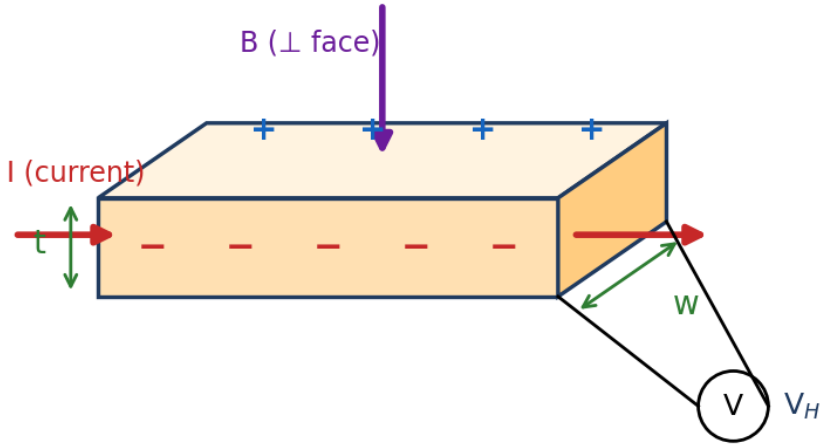
EXAMPLE • Numerical

In Si, $\mu_e = 0.135$ m²/Vs. $D_e = \mu kT/e = 0.135 \times 0.0259 =$ **3.5×10^{-3} m²/s**.

6 Hall Effect and its Applications

DEFINITION • Hall effect

When a current-carrying conductor or semiconductor is placed in a magnetic field perpendicular to the current, a transverse voltage (Hall voltage V_H) appears across the sample, perpendicular to both I and B.



n-type: electrons (drift opposite to I) are pushed to the front face by $F = -e(v \times B)$

Front face turns negative, back face positive → Hall voltage V_H across width w

Fig. 8: Hall effect in an n-type slab: current I along x, field B along z, Hall voltage along y

Carriers drifting with velocity v_d feel a sideways magnetic force Bev_d. They accumulate on one face until the electric Hall field E_H balances it: $eE_H = Bev_d$.

Hall field	$E_H = Bv_d = \frac{BJ}{ne} \quad (J = nev_d)$
Hall voltage (w = width, t = thickness, I = J w t)	$V_H = E_H w = \frac{IB}{net}$
Hall coefficient	$R_H = \frac{E_H}{JB} = \frac{V_H t}{IB} = -\frac{1}{ne}$ (n-type), $+\frac{1}{pe}$ (p-type)
Hall mobility	$\mu_H = \lvert R_H \rvert \sigma$

Sample	Majority carrier	Sign of R_H	Polarity of the front face (Fig.)
n-type	Electrons	Negative	Negative (electrons pile up)
p-type	Holes	Positive	Positive (holes pile up)
Metal (e.g. Cu)	Electrons	Negative, tiny (n ~ 10^{28})	V_H very small → poor Hall sensor

APPLICATIONS • Applications of the Hall effect

- **Type of semiconductor:** sign of V_H (or R_H) shows whether carriers are electrons (−) or holes (+).
- **Carrier concentration:** $n = 1/(e|R_H|)$.
- **Mobility:** $\mu = |R_H|\,\sigma$ using the measured conductivity.
- **Magnetic field measurement:** Hall probe (gaussmeter) since $V_H \propto B$.
- Hall sensors in position/speed sensing, current sensors, brushless motors, power measurement ($V_H \propto$ product of I and B).

EXAMPLE • Numerical

n-type sample, $n = 10^{22}$ m^{-3}, t = 0.5 mm, I = 10 mA, B = 0.5 T: $V_H = IB/(\text{net}) = (10^{-2} \times 0.5)/(10^{22} \times 1.6\times10^{-19} \times 5\times10^{-4}) =$ **6.25 mV**.
Given $R_H = 3.66\times10^{-4}$ m³/C and $\sigma = 112\ (\Omega m)^{-1}$: $n = 1/(eR_H) =$ **1.7×10^{22} m^{-3}**, $\mu = R_H\sigma =$ **0.041 m²/Vs**.

Mobility and temperature, direct and indirect gaps

Topic	Key point
Mobility vs T	Lattice scattering: $\mu \propto T^{-3/2}$ (dominant at high T). Impurity scattering: $\mu \propto T^{3/2}$ (dominant at low T). So μ peaks at some intermediate temperature.
Direct gap (GaAs, GaN)	CB minimum and VB maximum at the same k: electron–hole recombination gives a photon → LEDs and lasers.
Indirect gap (Si, Ge)	Minimum and maximum at different k: recombination needs a phonon too, so light emission is weak.
Compound semiconductors	III–V (GaAs, InP) and II–VI (ZnS, CdS) tune E_g and colour by composition.

EXAM CORNER • True or false?

- The Fermi level of an intrinsic semiconductor depends strongly on temperature. **False** (it stays near mid-gap).
- Doping an intrinsic semiconductor increases its conductivity. **True**.
- In an n-type semiconductor there are no holes. **False** ($p = n_i^2/N_d$, small but non-zero).
- Hall voltage is zero if B is parallel to I. **True** (no sideways force).

7 Worked Problems

EXAMPLE • 1. Intrinsic carriers

For Si (E_g = 1.12 eV) $n_i = 1.5\times10^{16}$ m^{-3} at 300 K. The ratio at 400 K is approximately $(400/300)^{3/2}\, e^{(E_g/2k)(1/300 - 1/400)} = 1.54 \times e^{6.5\times10^{3}\times8.33\times10^{-4}} = 1.54 \times e^{5.4} \approx$ **3.5×10^2**, so n_i(400 K) ≈ 5×10^{18} m^{-3}.

EXAMPLE • 2. Conductivity

n-type Si with $N_d = 10^{22}$ m^{-3}, μ_e = 0.135 m²/Vs: $\sigma \approx N_d e\mu_e = 10^{22} \times 1.6\times10^{-19} \times 0.135 =$ **216 $(\Omega m)^{-1}$**, about 5×10^5 times more than intrinsic Si (≈ 4.4×10^{-4}).

EXAMPLE • 3. Band gap from a photon

A GaAs LED emits λ = 870 nm: $E_g = 1240/870 =$ **1.43 eV**. Si (1.12 eV) absorbs light up to λ = 1240/1.12 = **1107 nm**.

EXAMPLE • 4. Fermi level of p-type

Si with $N_a = 10^{22}$ m^{-3} at 300 K: $E_i - E_F = kT \ln(N_a/n_i) =$ **0.347 eV**, so E_F is 0.347 eV below mid-gap (0.21 eV above E_v).

EXAMPLE • 5. Diffusion

Holes in Si with μ_h = 0.048 m²/Vs: $D_h = 0.048 \times 0.0259 =$ **1.24×10^{-3} m²/s**; with τ = 10 μs, $L_h = \sqrt{(D\tau)} =$ **1.1×10^{-4} m**.

Key terms at a glance

Term	Meaning
Generation / recombination	Creation / annihilation of an electron–hole pair
Majority / minority carriers	Carriers in excess / in deficit compared with the other type
Degenerate semiconductor	So heavily doped that E_F enters a band and it behaves like a metal
Mobility μ	Drift velocity per unit electric field
Carrier lifetime τ	Average time a minority carrier exists before recombining

EXAM CORNER • Try yourself (answers in brackets)

- Find n_i ratio for Ge when T rises from 300 K to 350 K (E_g = 0.67 eV). [≈ 6]
- A Si bar has $n = 5\times10^{21}$ m^{-3}, μ_e = 0.13. Find σ. [104 (Ωm)$^{-1}$]
- Hall coefficient of a sample is -8×10^{-5} m³/C. Find n and the carrier type. [7.8×10^{22} m^{-3}; electrons]
- Find V_H for I = 5 mA, B = 0.2 T, t = 0.2 mm, $n = 5\times10^{21}$ m^{-3}. [6.3 mV]

KEY IDEA • Problem-solving recipe

- **1.** Identify the type of material (intrinsic or n/p) and the temperature.
- **2.** Convert units: cm^{-3} → m^{-3} (×10^6), mm → m, eV → J (×1.6×10^{-19}); at 300 K kT = 0.0259 eV.
- **3.** Majority carrier ≈ doping level; get the minority carrier from $np = n_i^2$.
- **4.** Use $\sigma = e(n\mu_e + p\mu_h)$; locate E_F with $kT \ln(n/n_i)$ from E_i.

REMEMBER • Common mistakes

- Writing E_F at mid-gap for doped material (that is only for intrinsic).
- Forgetting that holes move along E while electrons move against E.
- Using the wrong sign for R_H of n-type material.
- Mixing units of mobility (cm²/Vs vs m²/Vs; 1 m²/Vs = 10^4 cm²/Vs).

8 Formula Sheet and Exam Corner

Topic	Key formulae
Intrinsic carriers	$n = N_c e^{-(E_c-E_F)/kT}$; $p = N_v e^{-(E_F-E_v)/kT}$; $np = n_i^2 = N_c N_v e^{-E_g/kT}$
Intrinsic E_F	$(E_c+E_v)/2 + (3kT/4)\ln(m_h^*/m_e^*)$
Conductivity	$\sigma = e(n\mu_e + p\mu_h)$; intrinsic: $\propto e^{-E_g/2kT}$
Extrinsic E_F	n-type: $(E_c+E_d)/2$ at 0 K; p-type: $(E_v+E_a)/2$ at 0 K
Currents	$J = e(n\mu_n + p\mu_p)E + e(D_n dn/dx - D_p dp/dx)$; $D/\mu = kT/e$
Hall	$V_H = IB/(net)$; $R_H = -1/ne$; $\mu = \lvert R_H\rvert\sigma$

EXAM CORNER • Short questions (2–3 marks)

- How are energy bands formed in solids?
- Distinguish between conductors, semiconductors and insulators using band diagrams.
- What is an intrinsic semiconductor? Where is its Fermi level?
- What are donor and acceptor impurities? Give examples.
- State the law of mass action.
- Define drift and diffusion currents.
- State Einstein's relation.
- Define Hall coefficient and write its expression.

EXAM CORNER • Long questions (10 marks)

- Derive expressions for the electron and hole concentrations in an intrinsic semiconductor and locate its Fermi level.
- Obtain an expression for the electrical conductivity of an intrinsic semiconductor and explain its temperature dependence.
- Derive the carrier concentration in n-type and p-type semiconductors. How does the Fermi level depend on doping and temperature?
- Explain drift and diffusion currents and derive Einstein's relation.
- Explain the Hall effect, derive the Hall coefficient and list its applications.

EXAM CORNER • Quick MCQs

- 1. Fermi level of an n-type semiconductor at 0 K lies (a) mid-gap (b) between E_d and E_c (c) below E_v.
- 2. Hall coefficient of a p-type semiconductor is (a) $-1/ne$ (b) $+1/pe$ (c) zero.
- 3. Resistivity of a semiconductor with rising T (a) increases (b) decreases (c) constant.
- 4. D/μ equals (a) kT/e (b) e/kT (c) kTe.
- **Answers:** 1-b, 2-b, 3-b, 4-a.

QUICK SUMMARY

- Atomic levels broaden into bands; the gap size separates conductors, semiconductors and insulators.
- Intrinsic: $n = p = n_i$, E_F at mid-gap, $\sigma \propto e^{-E_g/2kT}$. Extrinsic: doping fixes the majority carrier and moves E_F toward E_c (n) or E_v (p).
- Drift (field) and diffusion (gradient) carry current; $D/\mu = kT/e$. The Hall effect gives carrier type, density and mobility.